
ThousandWorlds: A benchmark for climate emulation of potentially habitable exoplanets

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Abstract

The search for life beyond Earth will depend on detecting faint signatures in the atmospheres of potentially habitable exoplanets. Interpreting those signatures requires understanding the host planet’s climate: the same molecule may signal life on one planet and abiotic chemistry on another. Global climate models (GCMs) provide this understanding, but individual runs can require millions of core-hours and substantial domain expert time. Machine-learning emulators could remove this bottleneck, but progress has been limited by the absence of a curated, multi-model exoclimate dataset. We introduce ThousandWorlds, an ML-ready benchmark for exoclimate emulation and for the broader regime of low-data, multi-simulator, parameter-to-field regression. The dataset contains approximately 1,700 simulations from five GCMs, mapping eight planet parameters to 3D atmospheric fields including temperature, humidity, winds, clouds, and radiation. Three nested subsets stage progressively harder challenges: single-simulator regression, multi-simulator regression with complete observations, and multi-simulator regression with structured missingness. We propose two evaluation protocols: one for ranking methods, and one that measures performance relative to the disagreement between GCMs themselves. We evaluate ten baselines spanning simple methods, trees, deep learning, and Gaussian processes. GP-based methods perform best, suggesting that ThousandWorlds exposes a regime where off-the-shelf deep learning does not yet succeed.

Data: withheld for double-blind review.

Code: withheld for double-blind review.

1 Introduction

We may be the first generation able to answer whether life exists beyond Earth. The most promising hosts, rocky habitable-zone planets, are common in the galaxy [Dressing and Charbonneau, 2015], and JWST has just begun observing the atmospheres of the nearest candidates. The search for life will hinge on the detection of *biosignatures* – molecular fingerprints of life imprinted as absorption lines in observed spectra. But these signatures are ambiguous – does an O_2 detection mean life, or just photodissociation of water [Wordsworth and Pierrehumbert, 2014]? Their interpretation requires knowledge of the planet’s climate. Temperature, circulation, clouds, and heat transport all matter, and predicting them accurately requires sophisticated 3D modelling.

This has driven an increasing use of *global climate models* (GCMs), large numerical codes that simulate 3D atmospheric fluid dynamics alongside non-dynamical processes like clouds and radiation. But a single GCM simulation typically costs 10^4 – 10^6 core-hours plus substantial domain expert time to configure and monitor, limiting studies to small ensembles of hand-picked configurations. An emulator that produces near-instant climate predictions would remove this bottleneck, opening the door to large parameter sweeps, principled uncertainty quantification, and integration with observational inference pipelines.

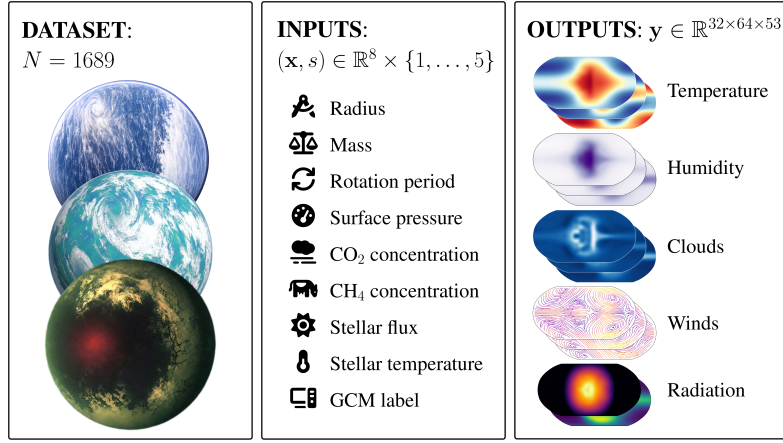


Figure 1: Dataset overview: inputs and the 53 output fields.

38 Despite this need, exoclimate emulation remains largely unexplored. The main barrier has been the
39 absence of a curated, multi-model dataset: the raw simulations exist, but they are scattered across
40 studies in incompatible formats, on different vertical grids, with different output variables.

41 Such a situation is not unique to exoclimate: across the sciences, many emulation problems share
42 the same difficult structure – a handful of input parameters, high-dimensional structured outputs,
43 scarce simulator evaluations, and several imperfect simulators. Existing scientific ML benchmarks
44 cover parts of this landscape, but largely target data-rich field-to-field prediction regimes where deep
45 learning succeeds [Takamoto et al., 2024, Ohana et al., 2025, Hu et al., 2026, Luo et al., 2024, Tali
46 et al., 2024], while the closely related multi-fidelity surrogate literature relies on synthetic or bespoke
47 test cases rather than shared community datasets [Fernández-Godino, 2023, Brunel et al., 2025]. The
48 complementary regimes of parameter-to-field regression, data scarcity, and multi-simulator learning
49 are comparatively overlooked.

50 We introduce ThousandWorlds, a benchmark that sits at the intersection of these three regimes
51 and fulfils the domain need for an ML-ready exoclimate dataset. Developed in collaboration with
52 exoplanet climate scientists, the dataset comprises approximately 1,700 simulations from five GCMs,
53 spanning totally ice-covered snowball worlds to steamy moist greenhouses. Each maps eight planet
54 parameters to 3D atmospheric variables covering temperature, humidity, winds, clouds, and radiation.
55 Three nested subsets stage increasingly demanding challenges: (1) single-simulator regression, (2)
56 multi-simulator transfer with complete observations, and (3) multi-simulator transfer with structured
57 missingness – the full dataset.

58 The closest domain precedents are Earth-system ML benchmarks [Rasp et al., 2024, Yu et al., 2024,
59 Cachay et al., 2021, Watson-Parris et al., 2022, Kaltenborn et al., 2023, Irvin et al., 2025]. One can
60 view ThousandWorlds as a benchmark within this climate modelling tradition, but with a different
61 source of diversity: rather than varying forcings on a single planet, the planet itself varies, producing
62 hugely diverse climate states by Earth-modelling standards. Within exoplanet astronomy, prior ML
63 work targets individual components within GCMs [Tahseen et al., 2024, Malsky et al., 2025] or hot
64 Jupiters: Plaschzug et al. [2025] train the only prior 3D exoclimate emulator, a pointwise model fit to
65 60 simulations from a single GCM, and Roth et al. [2024] provide the closest large-dataset precedent,
66 with 345 simulations. ThousandWorlds is one of few benchmarks to combine parameter-to-field
67 regression, multi-simulator transfer, and structured missingness in a single dataset, and the first large
68 ML-ready dataset for emulating potentially habitable exoplanet climates.

69 We define two evaluation protocols: the *standard* protocol for comparing methods, and the *shared-*
70 *planets* protocol, which measures emulator error against the inter-GCM spread, giving a clearer
71 measure of scientific utility. We evaluate ten baselines spanning simple methods, trees, deep learning,
72 and Gaussian processes (GPs). The GP methods prove to be the strongest baselines, suggesting that
73 this regime poses challenges for standard deep learning.

74 2 The ThousandWorlds dataset

75 2.1 Task description

76 **Planets.** We focus on tidally locked waterworlds (Figure 1): ocean-covered rocky planets with
77 one hemisphere permanently facing the host star. These planets are the most widely simulated

subclass of potentially habitable exoplanets. **Inputs.** Each planet is characterized by eight continuous inputs: radius, surface gravity, rotation period, surface pressure, CO₂ and CH₄ volume fractions, incident stellar flux, and stellar temperature. A discrete GCM label identifies the source GCM. We restrict evaluation to planets satisfying the *core domain* input constraints in Table 3. Beyond these bounds, physical plausibility and climate regime transitions would make the scientific interpretation of emulator accuracy less clear. However, we retain some simulations outside the core domain for training; these help anchor the response surface near the domain edges. **Outputs.** GCMs are run until the atmosphere reaches a statistical steady state, plus some averaging window; the prediction target is the mean over this window, represented as 53 fields on a 32×64 latitude–longitude grid. We use *variable* to refer to a single physical quantity that may span multiple vertical levels, and *field* to refer to a single slice at one level. The 3D variables are temperature, (specific) humidity, east–west (E–W) wind, north–south (N–S) wind, and cloud fraction, each on 10 pressure levels. The 2D variables are surface temperature, outgoing longwave radiation (OLR), and absorbed shortwave radiation (ASR).

2.2 Dataset construction

Our data spans five exoplanet GCMs. We designate two of them – ExoCAM and the UM – as *target* GCMs, evaluated at test time, since they are high-fidelity models with relatively plentiful simulations in the core domain. The remaining three serve as *auxiliary* GCMs, contributing training data only. The simulations are drawn partly from the existing literature and partly from our own bespoke runs, chosen to fill gaps in input-space coverage (Appendix A.1). We overview the dataset composition in Table 4 and the GCMs in Table 5.

2.3 Dataset challenges

Structured missingness. Different simulations use different vertical grids and do not all output the same variables. After interpolation onto the common 10-level pressure grid, this leaves a structured pattern of missingness: some simulations lack data at the uppermost or lowermost pressure levels, and some lack entire variables. We visualize the missingness patterns in Figure 2.

Inter-GCM disagreement. Different GCMs use different dynamical cores, physical parameterizations, and numerical methods. Two equal-fidelity GCMs given identical planet parameters can produce systematically different climates. This disagreement reflects genuine epistemic uncertainty about the atmospheric physics. Achieving positive transfer from shared cross-GCM structure (ideally reflecting the shared underlying physics) while avoiding negative transfer from their systematic differences is one of the benchmark’s key modelling challenges. The shared-planets evaluation protocol (Section 2.6) measures emulator error against this disagreement directly.

Within-GCM variability. Even within a single GCM, simulations from different studies use different configurations (e.g., convection schemes, cloud particle sizes, time-averaging windows, initial conditions). These choices affect the final climate but are numerous, inconsistently documented, and far too sparsely sampled to include as inputs. From an emulation perspective, they act as structured noise: output variability that correlates with unobserved simulation settings rather than with the eight planet parameters.

2.4 Data processing

Regridding. Each GCM produces output on its own native grid. We regrid all fields onto a 32×64 latitude–longitude grid with 10 pressure levels for 3D variables; the pressure levels are isobars defined relative to the input surface pressure. **Transformations.** Inputs and outputs are transformed to bring them onto ranges where squared error is well-behaved. **Spectral representation.** We release the dataset in two formats: gridded numpy arrays, the primary representation and the basis for all evaluation, and spectral coefficients after a (T21) spherical harmonic transform of each field. Further details are in Appendix A.3.

2.5 Benchmark structure

We define three nested subsets that progressively expose the challenges described in Section 2.3 (Table 6). **Single-complete** uses UM simulations only, with 48 fields common to all GCMs – a pure regression problem with no multi-GCM transfer and no missingness. **Multi-complete** adds all five GCMs but retains the same 48 fields, introducing cross-GCM transfer without missingness. **Multi-partial** uses all five GCMs and 53 fields (from a wider vertical grid), introducing structured missingness (top and bottom pressure levels and entire-variable absences); this is the full dataset. Train–test splitting is based on the 346 target simulations. All remaining simulations are training-only.

Table 1: RMSE by variable and method on the Multi-partial subset (the full dataset), with geometric means for the Multi-complete and Single-complete subsets in the final two rows. Darker colours indicate lower (better) scores, with bold the lowest in each row. Geometric means are reported relative to Train-mean’s. Scores are means over five training seeds. Table 11 gives per-variable results for all three subsets with standard deviations.

Variable	Train-mean	kNN	PCA-GBT	Coord-MLP	Coord-DeepONet	PCA-MLP	ConvDec	SFNO	PPCA-ICM	GPLFR
Surface temp. (K)	24.52	18.79	12.33	17.65	12.45	12.24	12.75	12.24	10.06	9.89
Temperature (K)	20.34	14.66	10.23	11.94	10.43	10.08	10.11	10.52	8.79	8.26
Humidity (dex)	1.119	0.784	0.536	0.627	0.527	0.5251	0.5253	0.531	0.446	0.432
Cloud fraction (1)	0.1277	0.0998	0.0879	0.1073	0.0897	0.0898	0.0816	0.0845	0.0836	0.0778
E–W wind (m s^{-1})	16.73	10.92	10.64	12.12	10.88	11.19	9.97	10.46	9.63	9.50
N–S wind (m s^{-1})	6.57	4.72	4.75	5.13	4.91	4.67	4.08	4.18	4.34	4.12
ASR (W m^{-2})	59.0	33.2	27.1	77.1	29.8	26.2	37.7	33.5	26.0	22.8
OLR (W m^{-2})	41.0	24.8	19.5	29.0	18.9	18.7	18.0	19.0	18.2	16.1
Geometric mean	1.00	0.685	0.550	0.754	0.560	0.546	0.547	0.551	0.492	0.462
Multi-complete geometric mean	1.00	0.700	0.540	0.723	0.554	0.542	0.546	0.548	0.513	0.458
Single-complete geometric mean	1.00	0.646	0.610	0.756	0.630	0.609	0.561	0.560	0.538	0.518

2.6 Metrics and evaluation protocols

Our primary benchmark metric is the area-weighted RMSE. For probabilistic methods, we also use the energy score, a proper scoring rule which can be seen as a multivariate generalization of the continuous ranked probability score (Appendix F.4). All metrics are computed per field and averaged over test examples; for 3D variables, we average uniformly over pressure levels. We explore additional metrics (anomaly correlation coefficient and spread–skill ratio) in Appendix F.

We evaluate under two protocols. The *standard* protocol reports metrics as usual on the full test sets. The *shared-planets* protocol restricts to planets simulated by both target GCMs. For each simulation in these same-planet pairs, the source GCM is treated as the ground truth and the other GCM as a competing predictor. Each metric is then reported as the ratio of the emulator’s score to the other GCM’s score. Values below 1 mean the emulator’s error is smaller than the disagreement between GCMs for the same planet; values above 1 mean it is larger. The shared-planets protocol applies only to the Multi- subsets, since it requires planets simulated by both target GCMs. The standard protocol has larger test sets and is more reliable for ranking methods; the shared-planets protocol puts errors on a scientifically meaningful scale.

2.7 Baselines

We evaluate ten baselines. Appendix B gives full descriptions and hyperparameter tuning details. **Train-mean** predicts the training-set mean for every test example. **kNN** averages the k nearest training examples in input space. **PCA-GBT** regresses PCA scores of the spectral coefficients (Section 2.4) with gradient-boosted trees. **Coord-MLP** follows Plaschzug et al. [2025]: a pointwise MLP maps planet parameters and a spatial coordinate to the field value at that coordinate. **Coord-DeepONet** applies a DeepONet architecture [Lu et al., 2021] to the same pointwise queries. **PCA-MLP** regresses the PCA scores of the spectral coefficients with an MLP. **ConvDec** is a convolutional decoder similar to the DCGAN generator of Radford et al. [2016], here trained as a deterministic regressor. **SFNO** adapts the Spherical Fourier Neural Operator of Bonev et al. [2023] to parameter-to-field regression. **PPCA-ICM** compresses outputs with probabilistic PCA and regresses the scores with a multi-task GP over GCM labels (ICM; Alvarez et al., 2012). **GPLFR** [Stevenson et al., 2026] replaces PPCA-ICM’s compress-then-predict pipeline with end-to-end optimization.

3 Experiments

3.1 Baseline comparison

Table 1 reports per-variable RMSE for the ten baselines on Multi-partial (and geometric means for the other two subsets). The two GP methods are the strongest baselines: GPLFR achieves the lowest RMSE on seven of the eight variables on the Multi- subsets, and three of eight on Single-complete, and PPCA-ICM is the second-best method on every subset by geometric-mean RMSE. Among the deep learning baselines, PCA-MLP, ConvDec, and SFNO are closely matched. The trees baseline PCA-GBT is competitive with the best deep learning methods, while Coord-MLP trails even kNN. Comparing to the simple baselines (Train-mean, kNN), the learned methods improve most clearly on temperature and humidity, and least on cloud fraction, winds, and ASR.

Table 2: Relative RMSE by variable and method under the shared-planets protocol on the Multi-partial subset, with the Multi-complete geometric mean in the final row. Scores are means over five training seeds. Table 12 gives per-variable results for both subsets with standard deviations.

Variable	Train-mean	kNN	PCA-GBT	Coord-MLP	Coord-DeepONet	PCA-MLP	ConvDec	SFNO	PPCA-ICM	GPLFR
Surface temp.	1.680	1.283	1.039	1.380	1.005	1.057	1.007	0.975	0.927	0.739
Temperature	1.655	1.209	0.925	0.998	0.949	0.902	0.903	0.921	0.829	0.664
Humidity	1.515	1.247	0.896	1.008	0.879	0.818	0.851	0.824	0.787	0.670
Cloud fraction	0.867	0.752	0.644	0.733	0.653	0.639	0.595	0.606	0.624	0.562
E-W wind	1.265	0.832	0.897	0.937	0.918	0.909	0.757	0.833	0.785	0.765
N-S wind	1.205	0.828	0.885	0.898	0.919	0.837	0.714	0.746	0.802	0.749
ASR	1.342	0.884	0.699	2.007	0.794	0.699	1.044	0.920	0.711	0.594
OLR	1.571	1.023	0.775	1.218	0.776	0.767	0.741	0.765	0.776	0.640
Geometric mean	1.360	0.988	0.836	1.097	0.854	0.819	0.814	0.816	0.776	0.669
Multi-complete geometric mean	1.365	1.040	0.809	1.061	0.862	0.816	0.827	0.823	0.822	0.688

Uncertainty in these RMSE numbers is dominated by the test-set draw rather than training seeds; Appendix F.2 quantifies this with paired bootstrap intervals. In Appendix D.2 we compare the two-stage compress-then-predict models (PCA-MLP and PPCA-ICM) with their end-to-end counterparts (Coord-DeepONet and GPLFR), which learn compression and regression jointly. Probabilistic evaluation and additional metrics are in Appendix F. Example predictions and climate diagnostics are in Appendix G.

The effect of auxiliary-GCM data is method-dependent: it slightly benefits PCA-GBT and GPLFR, slightly harms PCA-MLP and PPCA-ICM, and is clearly harmful for the remaining four deep learning methods (Coord-MLP, Coord-DeepONet, ConvDec, SFNO). Better ways to leverage multi-fidelity/multi-simulator data is therefore a promising direction for improving on these baselines. Data from outside the core domain is beneficial for all learned methods except PCA-GBT. See Appendix C.1 for further discussion.

3.2 Scientific utility of emulators

Under the shared-planets protocol, most learned baselines achieve relative RMSE below 1 on most variables (Table 2). So these emulators already reach accuracies within the spread of GCMs on these planets, while kNN sits right around the GCM-spread threshold. In Section 3.1 we saw that learned methods improved least over kNN on cloud fraction, winds, and ASR. Here we see that kNN already sits below the GCM disagreement on each of these variables (see Appendix D.1). Both probabilistic methods show a notable drop in relative energy score compared to relative RMSE (Appendix F.5).

For tasks where predictive accuracy matters more than physical consistency or interpretability – such as identifying promising observing targets or surveying habitability trends across parameter space – emulators operating within the GCM spread are likely already scientifically useful. That said, significant headroom remains: for example, our best method’s surface temperature RMSE (9.9 K) is large enough to affect habitability assessments, and many individual predictions do fall outside the GCM spread (Figure 3).

4 Conclusion

We have introduced ThousandWorlds, a benchmark for emulating exoplanet climates comprising approximately 1,700 simulations from five GCMs (limitations in Appendix E). On method comparison, our baseline results show that Gaussian-process-based methods outperform standard deep learning, making ThousandWorlds a useful challenge problem for novel deep learning surrogate methods in the low-data, parameter-to-field, and multi-simulator regime. GPLFR is the current best, with PPCA-ICM second on every subset; PCA-GBT and the deep learning methods PCA-MLP, ConvDec, and SFNO form a closely spaced group below them and are the natural targets for new methods. On scientific utility, the best baselines emulate individual GCM outputs more closely than GCMs agree with each other – a threshold of practical relevance for exoplanet astronomers – while leaving substantial headroom in absolute error terms.

We hope ThousandWorlds serves as both a useful benchmark problem for an underserved regime of scientific ML, and an invitation to the ML community to contribute their ideas and techniques to accelerate a field that aims to answer one of humanity’s oldest questions.

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A Dataset details

A.1 Sampling design

The bespoke simulations in Table 4 were selected using a weighted coverage design constructed by greedy sequential selection from a Sobol candidate pool. The design balances two competing goals: filling gaps in the eight-dimensional input space left by the heterogeneous literature data, and concentrating samples near the manifold of physically plausible planets. Pursuing coverage alone pushes towards combinations such as low radius with high surface gravity, producing unrealistically dense planets that often cause the GCMs to crash. The weighting function described below mitigates this by downweighting implausible regions.

We standardize the input space to $[0, 1]^8$ by recasting surface gravity as density ρ ; taking logarithms of radius, rotation period, surface pressure, CO_2 volume fraction, and CH_4 volume fraction; and rescaling the core-domain ranges in Table 3 to the unit cube. We then draw a scrambled Sobol candidate set U and an independent scrambled Sobol probe set U' , both in $[0, 1]^8$. Given an existing design $E \subset [0, 1]^8$ (the literature simulations, already standardized), we evaluate the weighted coverage objective

$$C_p(E \cup \{\mathbf{x}\}) = \int_{[0,1]^8} w(\mathbf{u}) r(\mathbf{u}; E \cup \{\mathbf{x}\})^p d\mathbf{u},$$

where $r(\mathbf{u}; E) = \min_{\mathbf{y} \in E} \|\mathbf{u} - \mathbf{y}\|_2$ is the distance from $\mathbf{u}$ to its nearest design point and $w(\mathbf{u})$ is the weighting function defined below. The integral is approximated by quadrature over the probe set U' . We use exponent $p = 2$, which biases selection towards shrinking large gaps without the brittleness of the minimax criterion ($p = \infty$). For a later batch of simulations we used $p = 4$ to more aggressively target remaining holes once a stable backbone of $p = 2$ samples was in place.

New points are selected greedily: at step t , we choose

$$\mathbf{x}_t = \arg \min_{\mathbf{x} \in U} C_p(E \cup \{\mathbf{x}_1, \dots, \mathbf{x}_{t-1}, \mathbf{x}\}).$$

The weighting function biases selection towards physically likely densities ρ given radius R , and physically likely rotation periods P_{rot} given stellar temperature T_* and incident flux F , leaving the remaining parameters unweighted. We set $w(\mathbf{x}) \propto w(\rho | R)w(P_{\text{rot}} | T_*, F)$, where:

- $w(\rho | R) = \text{Lognormal}(\rho; \mu_\rho, 0.15^2)$, with the mean set by the empirical radius–density relation for rocky planets from Müller et al. [2024]: $\mu_\rho / \text{g cm}^{-3} = 5.11 \times (R/R_\oplus)^{0.73}$.
- $w(P_{\text{rot}} | T_*, F) = \text{Lognormal}(P_{\text{rot}}; \mu_{P_{\text{rot}}}, 0.3^2)$. The mean $\mu_{P_{\text{rot}}}$ is computed from Kepler’s third law and the Stefan–Boltzmann law using a three-piece function that chains empirical stellar relations to obtain stellar mass M_* from T_* : for $T_* \in [2500, 3300]$ K, a radius–temperature relation from Cassisi and Salaris [2019], Stefan–Boltzmann to obtain luminosity L_* , and the mass–luminosity relation of Duric [2004]; for $T_* \in [3300, 4800]$ K, the mass–temperature and luminosity–temperature relations of Mann et al. [2013]; and for $T_* \in [4800, 5800]$ K, the mass–temperature and mass–luminosity relations of Moya et al. [2018].

For ExoPlaSim, we further restrict the sampling domain to $P_0, x_{\text{CO}_2} \leq 0.1$ bar and $x_{\text{CH}_4} = 0$, since ExoPlaSim is known to be unreliable outside this region [Paradise et al., 2022a]. A small number of

Table 3: Constraints on the eight continuous inputs that define the core domain.

Parameter (units)	Range
Radius (Earth radii)	[0.7, 1.75]
Surface gravity (m s^{-2})	[6.0, 17.0]
Rotation period (days)	[1, 200]
Surface pressure (bar)	[0.5, 5.0]
CO_2 volume fraction (%)	[0, 100]
CH_4 volume fraction (%)	[0, 5]
Incident stellar flux (W m^{-2})	[500, 1500]
Stellar temperature (K)	[2500, 5800]

Table 4: Dataset composition. ThousandWorlds contains 1,689 simulations. We restrict evaluation to 346 target simulations (bold); the remaining 1,343 are used for training only. These comprise 48 simulations from target GCMs that are outside the core domain (Table 3) and 1,395 simulations from auxiliary GCMs.

	GCMs	Physical domain		Data sources
		Core	Extended	
Target	UM	252	33	Mak et al. [2024], Sergeev et al. [2022], this work
	ExoCAM	94	15	Hammond et al. [2025], Haqq-Misra et al. [2022], Sergeev et al. [2022], Wolf et al. [2025], Woodward et al. [In preparation], this work
Auxiliary	ExoCAM-pre-2022	109	50	Komacek and Abbot [2019], kumar Kopparapu et al. [2016, 2017], Wolf et al. [2019], Wolf [2017], Suissa et al. [2020]
	LFRic	14	5	Haqq-Misra et al. [2022], this work
	ExoPlaSim	439	678	Macdonald et al. [2025], Paradise et al. [2021, 2022b], this work

Table 5: Overview of the five GCMs in ThousandWorlds. Further details in Appendix A.2.

GCM	Lineage	Key notes	Key reference(s)
ExoCAM	NCAR CESM/CAM adapted for exoplanets	Finite volume dynamical core; CAM4 cloud and convection physics; updated radiative transfer	[Wolf et al., 2022]
ExoCAM-pre-2022	Earlier versions of ExoCAM	Evolving radiative transfer schemes; known CO ₂ -atmosphere bias	[Yang et al., 2016, kumar Kopparapu et al., 2017, Wolf et al., 2019]
The UM	The UK Met Office’s Unified Model	Lat-lon grid; LLCs cloud scheme	[Boutle et al., 2017]
LFRic	The UM’s successor	Cubed-sphere grid	[Sergeev et al., 2023]
ExoPlaSim	PlaSim adapted for exoplanets	Simplified parameterizations; spectral dynamical core prone to Gibbs ringing	[Paradise et al., 2022a]

944 candidate points were discarded because the resulting simulations entered runaway greenhouse or
945 CO₂ atmospheric collapse.

946 A.2 GCM simulations

947 Below we describe the five GCMs used in ThousandWorlds (summarized in Table 5) alongside
948 the configurations used for our bespoke simulations. For simulations drawn from the literature,
949 configurations are documented in the source publications listed in Table 4, and the mapping from
950 individual simulations to source publications is provided in the dataset repository.

951 **ExoCAM.** ExoCAM is a planetary climate model based on National Center for Atmospheric
952 Research (NCAR) Community Earth System Model (CESM) version 1.2.1, which expands the native
953 capabilities to allow simulation of a broad parameter space of geophysical properties and atmospheric
954 compositions. ExoCAM operates as a patch for CESM. The user must first download the core CESM
955 code, which is freely available provided by NCAR¹, before modifying CESM with ExoCAM specific
956 configuration scripts, source code, and namelists, also freely available on GitHub². ExoCAM is
957 accompanied by a radiative transfer code ExoRT³, designed for flexibility across a broad range of
958 atmospheric compositions which is linked at build time. Wolf et al. [2022] describes ExoCAM,
959 ExoRT, and their relationship to CESM1.2.1.

960 ExoCAM provides basic controls for setting the geophysical and atmospheric properties of the
961 modelled planet, along with several options for controlling the modes of operation. ExoCAM permits
962 flexible setting of surface type (land vs. ocean), planet radius, surface gravity, rotation rate and
963 period, incident stellar flux, stellar spectrum, orbital eccentricity, obliquity, and partial pressure

¹<https://www.cesm.ucar.edu/models/cesm1.2/tags/>

²<https://github.com/storyofthewolf/ExoCAM>

³<https://github.com/storyofthewolf/ExoRT>

of atmospheric gases including N_2 , CO_2 , CH_4 , H_2O , O_3 , O_2 , H_2 , CO , NH_3 , along with cloud and aerosol species. ExoCAM is multi-configurable, allowing the user to configure with a variety of horizontal and vertical grid resolutions, along with different choices for dynamical core, convection, cloud, and aerosol physics. The simulations featured in this dataset uniformly use the finite volume (FV) dynamical core [Lin and Rood, 1996] along with CAM4 cloud and convection physics [Hack, 1994, Zhang and McFarlane, 1995, Rasch and Kristjánsson, 1998]. Likewise the simulations featured here use a horizontal resolution of $4^\circ \times 5^\circ$, but vertical levels used are mixed between 40 or 51 layers, representing 3 to 5 orders of magnitude extents in pressure space.

Post-2022 ExoCAM studies featured in the main training set are distinguished from early studies by a significant update to the radiative transfer component, ExoRT. Pre-2022 studies included in the auxiliary training set used an evolving set of radiative transfer configurations (e.g. Yang et al. [2016], kumar Kopparapu et al. [2017], Wolf et al. [2019]). By 2022, development had settled down to more-or-less its current form as described in Wolf et al. [2022].

UM. The Unified Model (UM) developed by the UK Met Office consists of the dynamical core, Even Newer Dynamics for General atmospheric modelling of the environment (ENDGame), which uses a semi-implicit semi-Lagrangian scheme to solve the non-hydrostatic, full deep-atmosphere equations of motion with varying gravity within the atmosphere [see Wood et al., 2014, Mayne et al., 2014a,b, for discussion]. The UM has been used to perform 3D climate simulation across a wide range of planets, ranging across the modern Earth [Walters et al., 2019, Andrews et al., 2020], rocky planets [Boutle et al., 2017, Sergeev et al., 2022, Mak et al., 2024] and gas giant exoplanets [Christie et al., 2022, Zamyatina et al., 2023, 2024, Mak et al., 2025]. The UM also consists of a 2-stream radiative transfer scheme, the “Suite of Community RAdiative Transfer codes based on Edwards and Slingo [1996] (Socrates), and uses the correlated- k method to solve for gaseous absorption from H_2O , CO_2 , O_3 , N_2O , CH_4 , O_2 , SO_2 , OCS from HITRAN [Rothman et al., 2013] and collision-induced absorption from $\text{N}_2\text{-N}_2$, $\text{N}_2\text{-CH}_4$, $\text{CO}_2\text{-CO}_2$ from HITRAN [Karman et al., 2019], and $\text{CH}_4\text{-CO}_2$ [Turbet et al., 2020]. Socrates is also used to construct the configuration file for the climate simulations which contains the optical properties of the gases in the shortwave (stellar) and longwave (planetary) part of the spectrum. The stellar spectrum is generated with the PHOENIX stellar model [Husser et al., 2013] and the star is assumed to have $\log(g) = 6.0$ and $[\text{Fe}/\text{H}] = 0.0$ for simplicity. The shortwave range spans $0.2\text{--}10\ \mu\text{m}$ and is binned into 6 bands, whereas the longwave range extends from $3.34\text{--}10^4\ \mu\text{m}$ and is binned into 9 bands.

The climate simulations are performed in a horizontal grid spacing of 2.5° in longitude and 2° in latitude, and a quadratically stretched vertical grid spacing of 38 layers to allow for higher resolution near the planetary surface. The vertical grid of the UM is altitude-based and the model domain height is fixed at 39.25 km across all simulations to maintain model stability. Lambert-Lewis (LLCS) simple moist adjustment scheme [Lambert et al., 2020] is used in the cloud treatment within the UM. All simulations are run for at least 20 Earth years to reach an equilibrium state. This is diagnosed by requiring that fluctuations in surface temperature and the top-of-atmosphere flux remain below 1% over the last 10 Earth years of the simulation time. The results presented in this work are temporally averaged over this final 10-year period.

LFRic-Atmosphere. LFRic-Atmosphere is the next-generation 3D GCM of the Met Office, designed for exascale computing [Adams et al., 2019, Bull et al., 2024]. This model combines on a new dynamical core GungHo and a suite of well-tested physical parameterizations inherited from its forerunner, the Unified Model (see previous section). Like the UM, GungHo solves the fully compressible non-hydrostatic Euler equations on a sphere but it uses a quasi-uniform cubed-sphere finite-element discretisation [Melvin et al., 2019, 2024]. For transport, the latest GungHo version uses a flux-form semi-Lagrangian scheme that ensures local conservation of mass and entropy, while maintaining other key properties of numerical schemes such as preservation of a constant, monotonicity and positivity [Bendall and Kent, 2025]. LFRic-Atmosphere is capable of reproducing a variety of atmospheric flows in different model configurations and domain geometries [Kent et al., 2023, Brown et al., 2024, Johnson et al., 2025], including idealised Earth-like exoplanet setups [Sergeev et al., 2023, 2024].

The LFRic-Atmosphere simulation used for these runs is derived from Sergeev et al. [2023] and based on the N_2 -dominated aquaplanet case of the TRAPPIST-1 Habitable Atmosphere Intercomparison [THAI, Fauchez et al., 2020, Sergeev et al., 2022]. The three parameters that we vary in this study are

(i) planet’s radius ($0.5\text{--}1.7R_{\text{T1e}}$), (ii) stellar flux ($0.6\text{--}1.7S_{\text{T1e}}$) and (iii) surface gravity ($0.9\text{--}1.15g_{\text{T1e}}$). We use the C24 cubed-sphere mesh resolution (i.e., $24 \times 24 \times 6$ atmospheric columns). In the vertical, the model uses the same vertical spacing as that in the UM: 38 quadratically stretched layers to allow for higher resolution near the surface. All simulations were run until a steady state.

ExoPlaSim. ExoPlaSim is an intermediate-complexity GCM that is a modification of the Planet Simulator (PlaSim) for planets beyond Earth. It has been used to study exoplanet habitability (e.g., Paradise et al., 2022a, Chen et al., 2023, Cohen et al., 2024, Macdonald et al., 2025) and for rocky planet parameter sweeps generally [Paradise et al., 2020, Macdonald et al., 2022]. ExoPlaSim has also been compared with higher complexity models like ExoCAM and found to replicate global climate patterns to first order [Paradise et al., 2022a]. However, ExoPlaSim lacks some of the advanced physics that higher complexity models have. For instance, it is prone to Gibbs oscillations when simulating tidally locked planets (a known problem in climate simulation more generally; Lander and Hoskins, 1997). For the simulations generated for this work, we use a T42 grid resolution and an exponential filter (the most consistent filter at low resolutions; Paradise et al., 2022a) to mitigate Gibbs oscillations.

A.3 Data processing details

Regridding. The 10 pressure levels are defined as relative isobars $\sigma_k = (P_k - P_{\text{top}})/(f_{\text{bottom}}P_0 - P_{\text{top}})$, where P_0 is the input surface pressure, $P_{\text{top}} = 10$ mbar, and $f_{\text{bottom}} = 0.925$ lifts the lowest level above near-surface pressure fluctuations. The σ_k are spaced according to a fourth-order polynomial that increases resolution near the top and bottom of the atmosphere. The 32×64 horizontal grid is a T21 Gaussian grid, supporting an exact spherical harmonic transform up to total wavenumber $\ell_{\text{max}} = 21$. Horizontal interpolation is bilinear; vertical interpolation is in log-pressure. Any field with partially missing values across the spatial grid (which can arise from pressure fluctuations in GCMs using height-based coordinates) is treated as fully unobserved.

Input and output transformations. Rotation period and surface pressure are log-transformed. Gas volume fractions are transformed via $x \leftarrow \text{asinh}(x/s)$ with a fixed pivot s per species (CO_2 : 10^{-6} ; CH_4 : 10^{-8}), chosen so that the transformation is approximately logarithmic at climatically significant concentrations and linear near zero. Humidity is log-transformed. Cloud fraction is smoothed-logit-transformed: $c \mapsto \text{logit}((c + \varepsilon)/(1 + 2\varepsilon))$ with $\varepsilon = 10^{-2}$; predictions are clamped back to $[0, 1]$ after inversion. ASR and OLR are divided by each example’s incident stellar flux.

Symmetrization. All planets in the dataset have symmetric forcing about the equator, so time-mean climates should be equatorially symmetric in all fields except N–S wind, which is antisymmetric. In practice, some simulations exhibit residual asymmetry due to finite time-averaging windows and, in some cases, spontaneous symmetry breaking or amplified numerical asymmetries. We treat these as artefacts (or at least beyond the scope of emulation) and symmetrize all fields in the released data: for symmetric fields this is equivalent to averaging the two hemispheres, and for the one antisymmetric field (N–S wind) to taking half the hemisphere difference.

A.4 Missingness patterns

Figure 2 shows every distinct observation pattern in the dataset. The dominant pattern is a missing top level: the native vertical grids of most ExoPlaSim simulations do not extend to 10 mbar, leaving the top level of each 3D variable unobserved in 826 of the 1,117 ExoPlaSim simulations. Entire-variable absences are generally studywise; for example, the 21 ExoCAM simulations from Hammond et al. [2025] do not include cloud fraction, ASR, or OLR. Overall, 5.8% of simulation–field pairs are unobserved.

A.5 Benchmark subset splits

Table 6 summarizes the three benchmark subsets (Section 2.5) and their split counts.

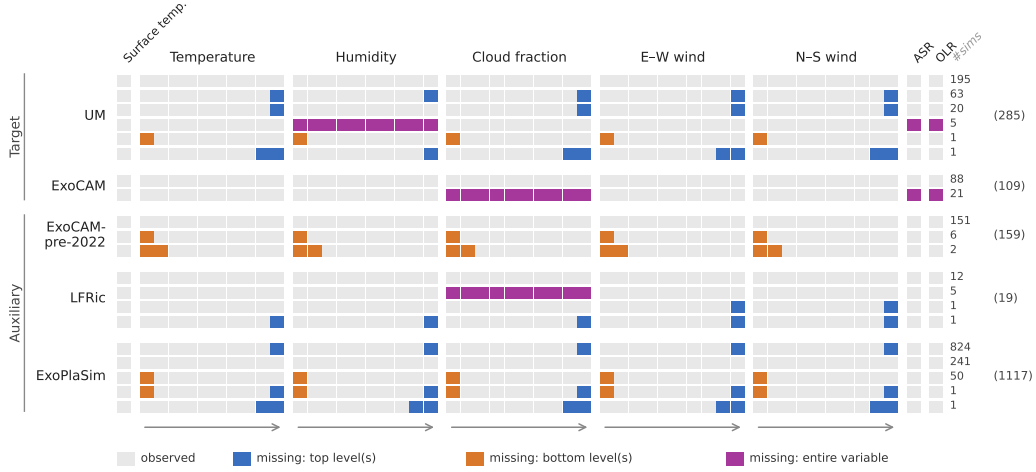


Figure 2: Observation patterns in the full dataset (Multi-partial). Each row is a distinct observation pattern occurring within one GCM, and the right-hand column counts the simulations sharing it. Columns are the 53 fields: three single-level variables, then five 3D variables on the common 10-level pressure grid ordered surface to top. Unobserved fields come in three kinds (Section 2.3): uppermost pressure levels (blue), lowermost pressure levels (orange), and entire variables (purple). Missingness is blockwise: a simulation either provides a field on the full spatial grid or not at all, and patterns are shared across simulator and study blocks rather than varying per example. The 48-field complete-observation subsets (Section 2.5) drop the top level of each 3D variable and exclude the simulations still missing any remaining field.

Table 6: Benchmark subsets: their features and split counts. Multi-partial is the full dataset; Multi-complete simplifies to complete observations only; Single-complete further simplifies to a single GCM (the UM). The shared-planets test sets are the restrictions of the standard test sets to planets simulated by both target GCMs (the UM and ExoCAM). To prevent leakage, we exclude simulations from auxiliary GCMs that correspond to identical planets present in the test set.

Subset	GCMs	Fields	Missingness	Train size	Standard test	Shared-planets test
Multi-partial	all 5 GCMs	53	yes	1527	128	64
Multi-complete	all 5 GCMs	48	no	1443	121	62
Single-complete	UM only	48	no	203	74	—

B Baseline details

B.1 Shared settings

Inputs and outputs. All baselines start from the transformed inputs and outputs described in Section 2.4. Inputs are then z-scored using training-set statistics. Output processing depends on the method family. Grid-space methods (Coord-MLP, Coord-DeepONet, ConvDec, SFNO) operate on the transformed gridded fields, z-scored per field using training-set statistics. Spectral-space methods (PCA-MLP, PCA-GBT, PPCA-ICM, GPLFR) first expand each horizontal field in the T21 spherical harmonic basis, then centre and scale per field, i.e., for field k , we compute the training-set mean $\bar{\mathbf{a}}^{(k)}$ and the root-mean anomaly energy

$$\sigma^{(k)} = \sqrt{\frac{1}{N} \sum_{i=1}^N \|\mathbf{a}_i^{(k)} - \bar{\mathbf{a}}^{(k)}\|_2^2}$$

and set $\mathbf{y}_i^{(k)} = (\mathbf{a}_i^{(k)} - \bar{\mathbf{a}}^{(k)}) / \sigma^{(k)}$. Dividing by $\sigma^{(k)}$ equalizes the total anomaly variance across fields. Predictions are denormalized by inverting these steps before evaluation. The simple baselines (training mean, kNN) operate directly on the transformed gridded fields without further standardization.

1077 **Equatorial symmetry.** The released targets are symmetrized as described in Appendix A.3. All
 1078 predictions are symmetrized in the same way before evaluation. For grid-space methods, symmetric
 1079 fields are averaged across hemispheres, except for N–S wind which is replaced by half the hemisphere
 1080 difference. For spectral-space methods, this can be done more neatly by zeroing the complementary
 1081 spherical harmonic coefficients (retaining $\ell + m$ even for symmetric fields, $\ell + m$ odd for the
 1082 antisymmetric N–S wind).

1083 **Output-space linear trend function.** The spectral-space methods subtract an affine function of the
 1084 planet parameters with per-GCM intercepts from each spectral coefficient before training, fit by ridge
 1085 regression (penalty 10^{-3}) on the training set and added back at prediction time.

1086 **Missing data handling.** Due to differences in GCM output grids, in the full dataset (Multi-partial),
 1087 some examples have unobserved output fields (Section 2.3). All methods handle this by restricting to
 1088 observed fields only. The training mean and kNN average fields only over examples where that field
 1089 is observed. Coord-MLP, Coord-DeepONet, ConvDec, and SFNO exclude missing fields from the
 1090 training loss. PCA-MLP, PCA-GBT, and PPCA-ICM implement PCA as probabilistic PCA (PPCA),
 1091 omitting missing fields from the likelihood. PCA-MLP and PCA-GBT retain only the point-estimate
 1092 scores, while PPCA-ICM combines PPCA noise with GP uncertainty for its ensemble predictions.
 1093 GPLFR handles missingness directly in its collapsed decoder likelihood by restricting each output
 1094 dimension’s contribution to only examples where it is observed.

1095 B.2 Models

1096 **Training mean.** The prediction for every test input is the per-field area-weighted mean over the
 1097 training set.

1098 **kNN.** For each test input, we find the k nearest training examples in standardized continuous input
 1099 space using Euclidean distance. GCM identity is encoded as a scaled one-hot vector appended to the
 1100 input, so that different GCMs contribute an additional distance penalty $\lambda\sqrt{2}$, encouraging same-GCM
 1101 neighbours while still permitting cross-GCM matches. The prediction is the uniform average of the
 1102 neighbours’ spatial output fields. We tuned k and λ (Table 7).

1103 **PCA-GBT.** Uses the same compress-then-predict pipeline as PCA-MLP – a shared linear trend
 1104 removed before PPCA compression of the normalized spectral coefficients – but replaces the MLP
 1105 with histogram-based gradient-boosted regression trees, fitting one ensemble per latent score from the
 1106 eight standardized planet parameters and a one-hot GCM label. Predictions are decoded through the
 1107 PCA loadings and means without noise sampling. We tuned the number of PCs, boosting learning rate
 1108 and maximum leaves per tree (Table 7); the remaining settings are fixed off-the-shelf (600 boosting
 1109 iterations with early stopping, a minimum of 10 samples per leaf, and L2 regularization of 1.0).

1110 **Coord-MLP.** This baseline adapts the pointwise emulator strategy of Plaschzug et al. [2025], who
 1111 trained a network to predict 3D temperature and wind fields for hot Jupiters from a 2D parameter
 1112 space of 60 GCM simulations. The model takes as input the concatenation of the eight standardized
 1113 planet parameters, a one-hot GCM label, a one-hot variable indicator, and the three spatial coordinates
 1114 (pressure level, latitude, longitude), and predicts a scalar field value. We tuned the network width,
 1115 depth, and learning rate (Table 7). The remaining model settings were fixed to Plaschzug et al.-style
 1116 choices: RMSProp optimizer, tanh activations, batch size 128, zero weight decay, trained with
 1117 unweighted pointwise MSE. The main adaptation relative to Plaschzug et al. is the input specification:
 1118 their model conditions on a 2D parameter space (stellar effective temperature and global-mean
 1119 temperature) with four output variables, whereas ours conditions on eight planet parameters with
 1120 GCM and variable identity tokens, predicting across all output fields. We also condition on a discrete
 1121 pressure-level coordinate rather than continuous pressure, since ThousandWorlds outputs are defined
 1122 on fixed relative isobars (Appendix A.3).

1123 **Coord-DeepONet.** Uses a DeepONet architecture [Lu et al., 2021], which models nonlinear
 1124 operators via an inner product between a branch network and a trunk network over a learned basis of
 1125 rank R . The branch takes the eight standardized planet parameters concatenated with a one-hot GCM
 1126 label. The trunk takes a one-hot variable indicator, normalized vertical level, latitude coordinate

1127 $\sin(\lambda)$, and periodic longitude features ($\sin \phi, \cos \phi$). The predicted value at a query point is

$$\hat{y}(\mathbf{x}, s, f, \ell, \lambda, \phi) = b_0(f, \ell, \sin \lambda, \sin \phi, \cos \phi) + \langle \mathbf{b}(\mathbf{x}, s), \mathbf{t}(f, \ell, \sin \lambda, \sin \phi, \cos \phi) \rangle / \sqrt{R},$$

1128 where $\mathbf{x}$ is the planet parameter vector, s the GCM label, f the variable, ℓ the vertical level, and
 1129 $\mathbf{b}$, $\mathbf{t}$, and b_0 are the branch, trunk, and bias networks respectively. All three are MLPs with SiLU
 1130 activations.

1131 Training uses scalar coordinate minibatches. Minibatches are sampled by first choosing a variable
 1132 uniformly, then a field within that variable (a single selection for single-level variables, a pressure
 1133 level for 3D variables), then a planet observed for that field, then latitude proportional to area weights
 1134 and longitude uniformly. For simplicity, branch, trunk, and bias MLPs share a single depth parameter.
 1135 We use AdamW with weight decay 10^{-4} and batch size 2^{15} . We tuned rank, branch and trunk widths,
 1136 shared depth, and learning rate (Table 7).

1137 **PCA-MLP.** Fits PCA on the normalized spectral coefficients to extract latent scores. (When fields
 1138 are missing, the PCA fit omits them from the likelihood, making it technically PPCA, but only point
 1139 estimates are retained.) A two-layer MLP (SiLU activations) then maps the inputs (eight standardized
 1140 planet parameters and a one-hot GCM label) to latent scores. We minimize MSE on the PCA scores
 1141 using AdamW. Predictions are decoded through the PCA loadings and means without noise sampling.
 1142 We tuned depth, number of PCs, hidden width (shared across layers), learning rate, and weight decay
 1143 (Table 7).

1144 **ConvDec.** A deterministic, parameter-conditioned convolutional decoder. A linear layer maps the
 1145 input vector (eight standardized planet parameters and a one-hot GCM label) to a 4×8 feature
 1146 grid with C channels, establishing the global spatial layout. Three decoder stages then each apply
 1147 $\times 2$ nearest-neighbour upsampling followed by 3×3 convolutions (SiLU activations), reaching the
 1148 full 32×64 resolution, with the channel count halved between successive stages; a final 1×1
 1149 convolution produces one channel per field (53 for Multi-partial, 48 for the complete subsets). This
 1150 coarse-to-fine design – project a vector to a small feature grid, then progressively upsample it with
 1151 shared convolutional filters – follows the DCGAN generator of Radford et al. [2016] (we borrow the
 1152 architecture only, training it by supervised regression rather than adversarially). We separate resizing
 1153 from convolution, with nearest-neighbour resizing, to avoid the uneven-overlap artifacts associated
 1154 with transposed convolutions [Odena et al., 2016]. Convolutions use circular padding in longitude
 1155 and zero padding in latitude. Training minimizes latitude-weighted MSE over observed fields using
 1156 AdamW with batch size 32. We tuned the seed channel count C , the number of convolutions per
 1157 stage, the learning rate, and the weight decay (Table 7).

1158 **SFNO.** Uses the Spherical Fourier Neural Operator of Bonev et al. [2023] via the
 1159 torch-harmonics reference implementation⁴ with its linear, zonal-kernel spectral filters: each
 1160 SFNO block applies a spherical harmonic transform, a learned complex channel-mixing matrix for
 1161 each degree ℓ shared across orders m , the inverse transform, and a pointwise GELU MLP with a
 1162 residual connection. Pointwise linear projections map the input channels to the embedding dimension
 1163 and back to one output channel per field. We use the native 32×64 Legendre–Gauss grid without
 1164 internal down-scaling and retain degrees $0 \leq \ell \leq 21$. We disable spatial normalization layers and the
 1165 input-residual path, and add a learned position embedding to the encoded features; these are explicit
 1166 configuration choices rather than reference defaults.

1167 To adapt this field-to-field architecture to parameter-to-field regression, we broadcast the inputs (eight
 1168 standardized planet parameters and a one-hot GCM label) as spatially constant channels, similar
 1169 in spirit to the FNO input-space extensions of Lu et al. [2022]. A constant field excites only the
 1170 $\ell = 0$ mode, and per-degree filters and pointwise operations preserve spatial constancy, so the learned
 1171 position embedding is the sole source of spatial symmetry breaking (spatial normalization would
 1172 remove precisely the $\ell = 0$ conditioning component, hence its exclusion). This anchoring to fixed
 1173 spatial coordinates is appropriate because the grids are substellar-fixed – the same property exploited
 1174 by Coord-MLP and Coord-DeepONet – though it means the full predictor, unlike the spectral filters
 1175 themselves, is not rotation-equivariant. Training follows ConvDec (latitude-weighted MSE over
 1176 observed fields, AdamW, batch size 32). We tuned the embedding dimension, number of blocks,
 1177 learning rate, and weight decay (Table 7).

⁴<https://github.com/NVIDIA/torch-harmonics>, version 0.8.0.

1178 **PPCA-ICM.** PPCA-ICM is a two-stage compress-then-predict pipeline on normalized spectral
 1179 coefficients.

1180 *Stage 1: PPCA compression.* PPCA compresses each example’s spectral coefficients to a latent score
 1181 vector, fit by EM. Missing fields are handled by omitting their likelihood terms.

1182 *Stage 2: ICM-GP regression.* The PPCA scores are regressed against inputs $(\mathbf{x}, s)$ using GPs with
 1183 an ICM kernel: $k((\mathbf{x}, s), (\mathbf{x}', s')) = k_x(\mathbf{x}, \mathbf{x}')B_{ss'}^{\text{in}}$, where k_x is an ARD kernel on the continuous
 1184 planet parameters and $\mathbf{B}^{\text{in}} \in \mathbb{R}^{S \times S}$ is a coregionalization matrix across GCMs; Table 7 gives the
 1185 selected kernel family for each subset. All scores share common lengthscales, amplitudes, and $\mathbf{B}^{\text{in}}$; a
 1186 per-component regime (separate lengthscales and amplitudes per score, with shared $\mathbf{B}^{\text{in}}$) performed
 1187 worse. Kernel hyperparameters are fit by maximizing the GP marginal likelihood using Adam.

1188 At prediction, ensemble members are generated by sampling scores from the GP predictive distri-
 1189 bution, decoding through the PPCA loadings and mean, and adding PPCA noise. Deterministic
 1190 predictions use the GP predictive mean decoded through the PPCA loadings. We tuned the number
 1191 of PCs, learning rate, and kernel family for each subset (Table 7).

1192 **GPLFR.** GPLFR is a GP-based model designed for high-dimensional structured outputs [Stevenson
 1193 et al., 2026]. It models each output as a linear decoding of a low-dimensional latent state drawn from
 1194 a GP prior over the inputs. The latents and GP kernel hyperparameters are jointly optimized under
 1195 a tempered MAP objective. In our configuration, GPLFR is essentially a more flexible, end-to-end
 1196 version of PPCA-ICM. It likewise uses an ARD kernel with ICM coregionalization over GCMs.
 1197 Optimization uses Adam with separate learning rates for the latent variables and global parameters.
 1198 Ensemble predictions are generated as in PPCA-ICM: sampling latents, decoding, and sampling
 1199 observation noise. We tuned latent dimensionality, regularization parameters (inverse-temperature
 1200 and latent noise), learning rates and kernel family (Table 7).

1201 B.3 Hyperparameter tuning

1202 We select hyperparameters via 5-fold cross-validation (CV). Only target simulations are assigned to
 1203 validation sets; all other training simulations (from auxiliary GCMs and/or outside the core domain)
 1204 are included in every fold’s training set. This applies to all methods except kNN, which keeps plain
 1205 k -fold CV (the target-only folds leak near-duplicate neighbours and collapse its k to 1). For each
 1206 candidate setting and fold, we evaluate RMSE (in normalized spectral-coefficient or grid-point space
 1207 depending on the method) on its validation set. We select the setting with the best mean performance
 1208 across folds, then refit on the full training set, using early stopping at the median best validation step
 1209 across folds. Tuned hyperparameters and search grids are shown in Table 7.

1210 For the learned methods, we search manually rather than exhaustively over the grid. We tune
 1211 primarily on the Multi-partial subset (the full dataset). We then do narrower retuning around these
 1212 Multi-partial-selected values for the Multi-complete subset, and a larger retune for Single-complete
 1213 (as it is significantly smaller).

1214 B.4 Compute resources

1215 All baselines except PCA-GBT were trained and evaluated on a single NVIDIA H100 GPU; ConvDec
 1216 and SFNO used H100 SXM5 GPUs, while the other GPU baselines used H100 PCIe GPUs. Table 8
 1217 reports wall-clock training times for the learned baselines on the Multi-partial subset (the full dataset).
 1218 Hyperparameter selection via 5-fold CV multiplies these costs by approximately $5 \times$ the number of
 1219 candidate settings evaluated per method (Appendix B.3). Train-mean and kNN require negligible
 1220 compute.

1221 C Ablations

1222 C.1 Dataset ablations

1223 We investigate two ablations of training set composition on Multi-partial using the standard test
 1224 set. For each ablation, we retuned the models’ key hyperparameters using the same CV protocol
 1225 as the main experiments (Appendix B.3; selected values in Table 7). The first ablation removes

Table 7: Hyperparameter search grids and selected values. Final steps were the median best validation steps across CV folds.

Hyperparameter	Subset(s)					Candidates
	Multi-partial	Multi-complete	Single-complete	Target-GCMs-only ablation	Target-only ablation	
kNN						
k	3	3	2	1	2	[1, 2, 3, 5, 10]
GCM penalty	1	1	0	1	10	[0, 0.3, 1, 3, 10]
PCA-GBT						
Number of PCs	150	150	50	50	50	[50, 150]
Learning rate	0.03	0.05	0.1	0.1	0.1	[0.03, 0.05, 0.1]
Max leaf nodes	63	31	15	15	15	[15, 31, 63]
Coord-MLP						
Width	1024	1024	512	1024	1024	[128, 256, 512, 1024]
Depth	4	4	6	4	4	[2, 4, 6]
Learning rate	$3 \cdot 10^{-4}$	$3 \cdot 10^{-4}$	$3 \cdot 10^{-4}$	$3 \cdot 10^{-4}$	$3 \cdot 10^{-4}$	$[3 \cdot 10^{-5}, 10^{-4}, 3 \cdot 10^{-4}, 0.001, 0.003]$
Final step	7800	7500	7900	7600	7500	—
Coord-DeepONet						
Rank	128	128	32	64	32	[32, 64, 128, 256]
Branch width	256	256	256	512	512	[64, 128, 256, 512]
Trunk width	512	512	256	512	512	[64, 128, 256, 512]
Depth	3	3	3	3	3	[2, 3]
Learning rate	0.003	0.003	0.003	0.003	0.003	$[3 \cdot 10^{-5}, 10^{-4}, 3 \cdot 10^{-4}, 0.001, 0.003]$
Final step	1500	1000	2000	4250	1500	—
PCA-MLP						
Number of PCs	50	50	20	50	50	[20, 50, 100, 150]
Depth	2	2	2	2	2	[1, 2]
Hidden widths	1024	1024	512	1024	1024	[32, 64, 128, 256, 512, 1024]
Learning rate	$3 \cdot 10^{-4}$	$3 \cdot 10^{-4}$	0.001	10^{-4}	10^{-4}	$[10^{-4}, 3 \cdot 10^{-4}, 0.001, 0.003, 0.01]$
Weight decay	10^{-4}	10^{-4}	10^{-4}	$3 \cdot 10^{-5}$	$3 \cdot 10^{-5}$	$[3 \cdot 10^{-5}, 10^{-4}, 3 \cdot 10^{-4}, 0.001, 0.003, 0.01, 0.03]$
Final step	600	500	300	900	700	—
ConvDec						
Seed channels	2048	2048	2048	2048	2048	[64, 128, 256, 512, 1024, 2048]
Convs per stage	1	1	2	2	2	[1, 2, 3]
Learning rate	10^{-4}	10^{-4}	10^{-4}	10^{-4}	10^{-4}	$[10^{-4}, 3 \cdot 10^{-4}, 0.001, 0.003, 0.01]$
Weight decay	0.3	0.3	0.3	0.3	0.1	$[3 \cdot 10^{-5}, 10^{-4}, 3 \cdot 10^{-4}, 0.001, 0.003, 0.01, 0.03, 0.1, 0.3]$
Final step	2700	2900	2100	2500	2500	—
SFNO						
Embedding dim.	768	768	768	768	512	[64, 128, 256, 384, 512, 768]
Number of blocks	2	2	2	4	4	[1, 2, 4, 6, 8]
Learning rate	$3 \cdot 10^{-4}$	$3 \cdot 10^{-4}$	0.001	0.001	0.001	$[10^{-4}, 3 \cdot 10^{-4}, 0.001, 0.003, 0.01]$
Weight decay	0.3	0.3	0.3	0.3	0.3	$[3 \cdot 10^{-5}, 10^{-4}, 3 \cdot 10^{-4}, 0.001, 0.003, 0.01, 0.03, 0.1, 0.3]$
Final step	2800	3000	2400	2800	2800	—
PPCA-ICM						
Kernel family	Matérn-3/2	Matérn-3/2	Matérn-3/2	Matérn-3/2	Matérn-3/2	[Matérn-3/2, Matérn-5/2, RBF]
Number of PCs	150	150	150	150	150	[20, 50, 100, 150]
Learning rate	0.003	0.003	0.001	0.03	0.03	$[3 \cdot 10^{-4}, 0.001, 0.003, 0.01, 0.03]$
Final step	620	600	960	60	40	—
GPLFR						
Kernel family	Matérn-5/2	Matérn-5/2	Matérn-3/2	Matérn-3/2	Matérn-3/2	[Matérn-3/2, Matérn-5/2, RBF]
Latent dimensionality	150	150	150	150	150	[50, 100, 150]
Inverse-temperature	0.1	0.1	0.03	0.03	0.03	[0.01, 0.03, 0.1]
Latent noise	0.10	0.03	0.03	0.03	0.03	[0.01, 0.03, 0.1]
Latent learning rate	0.1	0.1	0.1	0.1	0.1	[0.03, 0.1, 0.3]
Global learning rate	0.3	0.3	0.3	0.1	0.1	[0.03, 0.1, 0.3]
Final step	115	120	30	60	60	—

Table 8: Approximate wall-clock training time per run on the Multi-partial subset. PCA-GBT is trained on 32 CPU cores; all other methods use a single H100 GPU.

Method	Training time (minutes)
PCA-GBT	2.7
Coord-MLP	0.4
Coord-DeepONet	0.9
PCA-MLP	0.2
ConvDec	1.8
SFNO	2.4
PPCA-ICM	1.3
GPLFR	1.2

all auxiliary-GCM simulations (ExoPlaSim, LFRic, ExoCAM-pre-2022), reducing (Multi-partial) training from 1,527 to 266 target-GCM-only simulations. The second additionally removes the remaining 48 extended-domain simulations, leaving 218 core-domain-only simulations. (See Table 4 for a reminder of the dataset composition.)

Effect of auxiliary GCM data (Table 9). The effect of auxiliary GCM data is method-dependent: PCA-GBT and GPLFR benefit (by 2.0% and 0.1% in mean seed-0 RMSE), PPCA-ICM and PCA-MLP are roughly neutral (−2.2% and −1.6%), and the remaining methods are harmed, most strongly SFNO (−13.8%), Coord-DeepONet (−12.8%), and Coord-MLP (−11.7%). So it is plausible that the auxiliary set’s outputs are systematically different enough from those of the target GCMs to be causing negative transfer. To see if we could reduce the harm caused by auxiliary GCM data for the strongly affected deep learning models (ConvDecoder, SFNO, Coord-MLP, and Coord-DeepONet), we tried subtracting the same output-space linear trend function with per-GCM intercepts that the spectral-space methods used (Appendix B.1) before training, but this only had a small, inconsistent effect on their RMSEs (−2.6% to +1.5% in geometric-mean RMSE), and the auxiliary-GCM harm persists (−7.8% to −14.6%). The negative transfer seen is therefore not explained by these per-GCM offsets. These findings suggest that better multi-fidelity/multi-simulator transfer methods are a promising direction for improving on our baselines – particularly for those four deep learning methods.

Effect of extended-domain data (Table 10). The 48 extended-domain target-GCM simulations constitute around 18% of the target-GCM-only training set. Despite their small number, they are consistently beneficial for the learned methods: removing them increases RMSE for vast majority of variables across the learned methods. These simulations often sit near the boundaries of the core domain and may be helping anchor predictions there.

Table 9: Auxiliary GCM data ablation results on the Multi-partial subset. Each entry is the percentage change in RMSE when auxiliary GCM simulations are removed from the training set. Positive values indicate auxiliary data was helping; negative values indicate it was hurting. Results are reported for training seed 0.

Variable (%)	Train-mean	kNN	PCA-GBT	Coord-MLP	Coord-DeepONet	PCA-MLP	ConvDec	SFNO	PPCA-ICM	GPLFR
Surface temp.	−3.7	−14.5	+4.9	−23.4	−16.9	−3.1	−18.5	−20.6	−5.5	−3.0
Temperature	−3.3	−4.3	+6.4	−27.1	−15.6	−2.6	−16.5	−22.1	−2.9	+3.7
Humidity	−0.8	−14.2	+7.2	−23.7	−13.3	+3.8	−16.3	−17.9	+3.8	+2.3
Cloud fraction	−6.2	−3.7	+2.7	+5.1	−9.1	+2.5	−10.5	−10.3	−3.2	−1.4
E-W wind	−4.4	−3.8	−2.7	−11.5	−5.5	−0.6	−10.6	−11.0	+0.7	−1.8
N-S wind	−7.2	−5.8	−6.7	−6.2	−16.2	−1.3	−4.6	−8.4	−0.5	−5.3
ASR	−25.0	−6.3	+0.3	+14.2	−14.1	−6.5	−2.4	+0.6	−6.3	+4.6
OLR	−18.9	−10.3	+4.0	−21.0	−11.9	−4.6	−6.0	−15.7	−3.6	+1.9
Mean	−8.7	−7.9	+2.0	−11.7	−12.8	−1.6	−10.7	−13.2	−2.2	+0.1

Table 10: Extended-domain data ablation. Each entry is the percentage change in RMSE when the 48 target-GCM simulations outside the core domain are additionally removed from the target-GCM-only training set. Positive values indicate that these extended-domain data were helping. Results are reported for training seed 0.

Variable (%)	Train-mean	kNN	PCA-GBT	Coord-MLP	Coord-DeepONet	PCA-MLP	ConvDec	SFNO	PPCA-ICM	GPLFR
Surface temp.	+0.3	-3.8	-0.2	+22.5	+8.0	+17.1	+10.6	+10.7	+15.8	+14.5
Temperature	-0.8	-6.2	+1.0	+12.9	+5.9	+14.3	+5.4	+12.5	+10.4	+10.9
Humidity	-2.2	-4.0	-1.1	+5.2	+12.2	+9.6	+4.6	+9.4	+9.3	+8.6
Cloud fraction	-0.0	-5.9	-1.7	-6.7	+7.8	+0.7	+3.4	+7.2	+2.5	+7.0
E-W wind	-0.2	-0.3	+1.7	+2.4	+2.6	+5.7	+4.9	+4.5	-0.4	+5.2
N-S wind	-0.1	-1.4	+3.8	+0.1	+9.3	+5.2	+1.9	+1.2	+0.4	+5.4
ASR	+0.5	-10.5	+0.5	+2.6	+13.3	-1.6	+18.5	-1.7	+7.1	+5.0
OLR	-1.3	-6.5	-4.1	-6.4	+7.5	-2.8	+10.5	+5.5	+2.6	+4.8
Mean	-0.5	-4.8	-0.0	+4.1	+8.3	+6.0	+7.5	+6.2	+6.0	+7.7

D Additional discussion

D.1 Why is kNN strong on clouds, winds, and ASR?

In Section 3.1 we noted that kNN is relatively weak on temperature fields and humidity, but competitive with learned methods on cloud fraction, winds, and ASR. Taking winds as an example, kNN is likely strong because global wind patterns are quite spatial-template-like: nearby planets often share the same broad circulation regime, so local averaging already captures much of the learnable variation. The remaining wind errors may then be dominated by circulation regime transitions (see, e.g., Haqq-Misra et al., 2018) or spatial shifts of wind features like jets or vortices, which are harder to learn than the smoother global trends of temperature fields, for example, particularly under L_2 -based objectives [Subich et al., 2025]. Clouds and ASR (after removing the fixed insolation geometry component) are also relatively regime-determined versus trend-determined [Yang et al., 2013], and so kNN could be strong on these variables for the same reason. Cloud fraction is additionally highly dependent on both GCM and parameterization choices (as suggested by the low relative RMSE of Train-mean in Table 2; see also, e.g., Sergeev et al., 2022), limiting any cross-GCM positive transfer that might otherwise help the learned methods. (This is consistent with the long-standing difficulty of modelling clouds. For Earth, where we have better ground-truth access, clouds are usually cited as the largest source of uncertainty for climate change predictions; Sherwood et al., 2020.)

D.2 Learning compression and regression separately or jointly

Two pairs of baselines can be viewed as differing primarily in whether they learn their output basis (compression) and input-to-latent map (regression) separately or jointly: PCA-MLP versus Coord-DeepONet, and PPCA-ICM versus GPLFR (see Lu et al., 2022 and Stevenson et al., 2026 for these perspectives). The *separate* (two-stage) approach is simpler and more stable but can waste capacity on high-variance output structure that is irrelevant or hard to predict. The *joint* approach can better allocate capacity to predictable structure but pays for this flexibility with weaker identifiability, which can manifest as harder optimization or lower data efficiency. Our results appear consistent with this interpretation: both joint methods improve relative to their two-stage counterpart when moving from Single-complete to the larger Multi- subsets. However, which approach wins overall differs between the two families: in the deep learning pair, the two-stage method (PCA-MLP) leads on all three subsets; in the GP pair, the joint method (GPLFR) does. This is plausibly due to the stronger inductive biases of the GP methods allowing the joint learning advantages to win out at even at small dataset sizes.

E Limitations

Scope. ThousandWorlds restricts attention to tidally locked waterworlds (*aquaplanets*). Two facts make this the natural class to study. First, most detectable habitable-zone planets orbit close to stars dimmer than the Sun, where stronger tidal forces synchronize the planet’s rotation with its

orbit, locking one side in permanent daylight and the other in permanent night.⁵ Second, many such planets are likely to be ocean-covered, and a global ocean provides a clean idealization that avoids arbitrary choices about continent arrangement. These planets are the most widely simulated subclass of potentially habitable exoplanets, but still only a slice of the broader parameter space (e.g., dry planets, eccentric orbits, asynchronous rotators are excluded). Extending the benchmark to other planet classes is a clear avenue for future work.

GCMs. The auxiliary-GCM set consists primarily of ExoPlaSim (1,117 of 1,295 simulations; Table 4), which is an intermediate-complexity GCM (Appendix A.2). These simulations are likely less useful for transfer than the same number of high-fidelity GCM runs would be, although they can still be beneficial for the right model (see Appendix C.1). Our four high-fidelity GCMs share substantial heritage, particularly the UM with LFRic, and ExoCAM with ExoCAM-pre-2022 (see Appendix A.2 for detail). The shared-planets protocol evaluates one GCM from each lineage (the UM and ExoCAM), so it captures inter-lineage disagreement, but this is still only an estimate of the true epistemic uncertainty across exoplanet GCM space (e.g., as reported for a select planet by the THAI intercomparison; Sergeev et al., 2022).

Test-set sizes. Test sets are small (up to 128 simulations for the largest subset, Multi-partial). Fine-grained method comparisons should therefore be interpreted cautiously; see Appendix F.2 for estimates of score uncertainty due to test-set composition.

F Additional results

F.1 Result tables with seed variability

Tables 11 and 12 repeat the main RMSE and relative RMSE tables with standard deviations over random seeds.

F.2 Bootstrap intervals for Multi-partial RMSE results

Table 13 reports 95% bootstrap confidence intervals on the Multi-partial RMSE results for training seed 0. The intervals are notably wider than the training-seed standard deviations in Appendix F.1, indicating that test-set composition is the dominant source of uncertainty in the RMSE numbers.

A paired bootstrap test shows the method ranking more clearly under this uncertainty: within each resample, we compare the geometric-mean RMSE across the eight variables for each adjacently ranked pair of methods (Table 14). GPLFR’s lead is robust, ahead of PPCA-ICM in 99.5% of resamples, and PPCA-ICM is ahead of ConvDec in every resample. The methods from ConvDec down to Coord-DeepONet are closely matched, with adjacent orderings reversing in 22–42% of resamples. Coord-DeepONet is then clearly ahead of kNN, which is in turn clearly ahead of Coord-MLP, the weakest learned method.

F.3 Anomaly correlation coefficient (ACC) results

ACC measures agreement in spatial structure between prediction $\hat{\mathbf{y}}$ and truth $\mathbf{y}$, ignoring differences in global mean and amplitude. For a single field,

$$\text{ACC}(\hat{\mathbf{y}}, \mathbf{y}) = \frac{\langle \hat{\mathbf{y}}^\circ, \mathbf{y}^\circ \rangle_G}{\|\hat{\mathbf{y}}^\circ\|_G \|\mathbf{y}^\circ\|_G}, \quad \mathbf{u}^\circ \equiv \mathbf{u} - \bar{u}\mathbf{1}, \quad \bar{u} = \langle \mathbf{1}, \mathbf{u} \rangle_G / \langle \mathbf{1}, \mathbf{1} \rangle_G,$$

where $\langle \mathbf{u}, \mathbf{v} \rangle_G \equiv \mathbf{u}^\top \mathbf{G} \mathbf{v}$ is the area-weighted inner product. $\text{ACC} = 1$ indicates perfect spatial correlation; for a random prediction, $\mathbb{E}[\text{ACC}] = 0$.

GPLFR achieves the highest ACC on average across variables and subsets, followed by PPCA-ICM, kNN, PCA-GBT, and SFNO at roughly similar levels (Table 15). kNN’s relative strength on ACC, particularly winds and humidity, compared to RMSE suggests that spatial patterns are more locally consistent in input space than global means and amplitudes. Cloud fraction, N–S wind, and temperature have the most challenging spatial patterns as measured by ACC. All models achieve very

⁵This is why the Moon always shows the same face to Earth.

Table 11: RMSE by subset, variable, and method, reported as mean $\pm$ standard deviation over random seeds. Deterministic baselines are shown without standard deviations. The main-text Table 1 shows only the means.

Sub-set	Variable	Train-mean	kNN	PCA-GBT	Coord-MLP	Coord-DeepONet	PCA-MLP	ConvDec	SFNO	PPCA-ICM	GPLFR
Multi-partial	Surface temp. (K)	24.52	18.79	12.33 $\pm$ 0.48	17.65 $\pm$ 1.55	12.45 $\pm$ 0.64	12.24 $\pm$ 0.30	12.75 $\pm$ 0.68	12.24 $\pm$ 0.62	10.06 $\pm$ 0.01	9.89$\pm$0.07
	Temperature (K)	20.34	14.66	10.23 $\pm$ 0.11	11.94 $\pm$ 0.97	10.43 $\pm$ 0.49	10.08 $\pm$ 0.16	10.10 $\pm$ 0.27	10.52 $\pm$ 0.21	8.79 $\pm$ 0.02	8.26$\pm$0.09
	Humidity (dex)	1.119	0.784	0.536 $\pm$ 0.013	0.627 $\pm$ 0.061	0.527 $\pm$ 0.026	0.525 $\pm$ 0.008	0.525 $\pm$ 0.012	0.531 $\pm$ 0.011	0.446 $\pm$ 0.002	0.432$\pm$0.004
	Cloud fraction (I)	0.1277	0.0998	0.0879 $\pm$ 0.0007	0.1073 $\pm$ 0.0024	0.0897 $\pm$ 0.0013	0.0898 $\pm$ 0.0009	0.0816 $\pm$ 0.0018	0.0845 $\pm$ 0.0012	0.0836 $\pm$ 0.0004	0.0778$\pm$0.0003
	E-W wind (m s $^{-1}$)	16.73	10.92	10.64 $\pm$ 0.07	12.12 $\pm$ 0.28	10.88 $\pm$ 0.18	11.19 $\pm$ 0.04	9.97 $\pm$ 0.45	10.46 $\pm$ 0.25	9.63 $\pm$ 0.01	9.50$\pm$0.06
	N-S wind (m s $^{-1}$)	6.57	4.72	4.75 $\pm$ 0.04	5.13 $\pm$ 0.32	4.91 $\pm$ 0.09	4.67 $\pm$ 0.03	4.08$\pm$0.13	4.18 $\pm$ 0.07	4.34 $\pm$ <0.01	4.12 $\pm$ 0.01
	ASR (W m $^{-2}$)	59.0	33.2	27.1 $\pm$ 0.1	77.1 $\pm$ 9.7	29.8 $\pm$ 1.0	26.2 $\pm$ 0.7	37.7 $\pm$ 1.2	33.5 $\pm$ 1.0	26.0 $\pm$ 0.1	22.8$\pm$0.2
	OLR (W m $^{-2}$)	41.0	24.8	19.5 $\pm$ 0.4	29.0 $\pm$ 3.0	18.9 $\pm$ 0.2	18.7 $\pm$ 0.4	18.0 $\pm$ 0.7	19.0 $\pm$ 0.3	18.2 $\pm$ 0.1	16.1$\pm$0.1
	Geometric mean	1.00	0.685	0.550 $\pm$ 0.005	0.752 $\pm$ 0.023	0.560 $\pm$ 0.012	0.546 $\pm$ 0.006	0.547 $\pm$ 0.013	0.551 $\pm$ 0.007	0.492 $\pm$ 0.001	0.462$\pm$0.002
	Geometric mean	1.00	0.685	0.550 $\pm$ 0.005	0.752 $\pm$ 0.023	0.560 $\pm$ 0.012	0.546 $\pm$ 0.006	0.547 $\pm$ 0.013	0.551 $\pm$ 0.007	0.492 $\pm$ 0.001	0.462$\pm$0.002
Multi-complete	Surface temp. (K)	24.00	19.58	12.59 $\pm$ 0.30	16.66 $\pm$ 0.78	12.07 $\pm$ 1.27	12.36 $\pm$ 0.23	12.75 $\pm$ 0.46	12.29 $\pm$ 0.72	11.20 $\pm$ 0.01	9.79$\pm$0.08
	Temperature (K)	19.32	15.22	9.90 $\pm$ 0.16	11.44 $\pm$ 1.15	9.68 $\pm$ 0.62	9.47 $\pm$ 0.36	9.97 $\pm$ 0.23	10.40 $\pm$ 0.40	9.59 $\pm$ <0.01	8.14$\pm$0.04
	Humidity (dex)	1.087	0.765	0.506 $\pm$ 0.005	0.620 $\pm$ 0.029	0.490 $\pm$ 0.020	0.506 $\pm$ 0.009	0.512 $\pm$ 0.018	0.513 $\pm$ 0.035	0.459 $\pm$ <0.001	0.404$\pm$0.002
	Cloud fraction (I)	0.1356	0.1074	0.0913 $\pm$ 0.0005	0.1117 $\pm$ 0.0029	0.0968 $\pm$ 0.0020	0.0936 $\pm$ 0.0005	0.0865 $\pm$ 0.0007	0.0890 $\pm$ 0.0013	0.0878 $\pm$ <0.0001	0.0816$\pm$0.0003
	E-W wind (m s $^{-1}$)	15.25	10.16	8.92 $\pm$ 0.13	10.14 $\pm$ 0.13	9.63 $\pm$ 0.27	9.57 $\pm$ 0.09	8.64 $\pm$ 0.17	9.13 $\pm$ 0.25	8.63 $\pm$ <0.01	8.24$\pm$0.05
	N-S wind (m s $^{-1}$)	6.15	4.90	4.43 $\pm$ 0.07	4.89 $\pm$ 0.12	4.87 $\pm$ 0.19	4.66 $\pm$ 0.04	4.00$\pm$0.09	4.12 $\pm$ 0.15	4.22 $\pm$ <0.01	4.07 $\pm$ 0.03
	ASR (W m $^{-2}$)	59.7	31.5	26.5 $\pm$ 0.4	68.6 $\pm$ 7.7	29.6 $\pm$ 1.0	25.5 $\pm$ 0.5	36.5 $\pm$ 0.5	31.7 $\pm$ 1.7	26.1 $\pm$ <0.01	22.1$\pm$0.1
	OLR (W m $^{-2}$)	41.2	23.8	19.0 $\pm$ 0.3	26.2 $\pm$ 0.9	18.2 $\pm$ 0.5	18.5 $\pm$ 0.2	17.5 $\pm$ 0.1	18.3 $\pm$ 0.7	18.2 $\pm$ <0.1	15.5$\pm$0.1
	Geometric mean	1.000	0.700	0.540 $\pm$ 0.004	0.722 $\pm$ 0.025	0.554 $\pm$ 0.018	0.542 $\pm$ 0.006	0.546 $\pm$ 0.007	0.548 $\pm$ 0.012	0.513 $\pm$ <0.001	0.458$\pm$0.002
	Geometric mean	1.000	0.700	0.540 $\pm$ 0.004	0.722 $\pm$ 0.025	0.554 $\pm$ 0.018	0.542 $\pm$ 0.006	0.546 $\pm$ 0.007	0.548 $\pm$ 0.012	0.513 $\pm$ <0.001	0.458$\pm$0.002
Single-complete	Surface temp. (K)	20.85	14.04	11.63 $\pm$ 0.41	14.76 $\pm$ 1.17	11.94 $\pm$ 0.40	11.40 $\pm$ 0.23	10.72 $\pm$ 0.28	10.04 $\pm$ 0.44	9.65 $\pm$ <0.01	9.38$\pm$0.10
	Temperature (K)	18.71	11.71	9.88 $\pm$ 0.09	9.85 $\pm$ 1.00	9.75 $\pm$ 0.30	9.72 $\pm$ 0.09	8.64 $\pm$ 0.19	8.55 $\pm$ 0.08	8.33 $\pm$ <0.01	8.08$\pm$0.07
	Humidity (dex)	1.067	0.583	0.555 $\pm$ 0.016	0.582 $\pm$ 0.048	0.549 $\pm$ 0.024	0.523 $\pm$ 0.006	0.477 $\pm$ 0.013	0.472 $\pm$ 0.009	0.455 $\pm$ <0.001	0.447$\pm$0.003
	Cloud fraction (I)	0.1287	0.0902	0.0866 $\pm$ 0.0010	0.1021 $\pm$ 0.0035	0.0896 $\pm$ 0.0016	0.0900 $\pm$ 0.0007	0.0761$\pm$0.0004	0.0809 $\pm$ 0.0013	0.0816 $\pm$ <0.0001	0.0767$\pm$0.0003
	E-W wind (m s $^{-1}$)	12.75	8.53	8.54 $\pm$ 0.12	9.01 $\pm$ 0.27	9.55 $\pm$ 0.66	9.39 $\pm$ 0.09	7.34$\pm$0.05	8.16 $\pm$ 0.14	8.14 $\pm$ <0.01	7.49 $\pm$ 0.07
	N-S wind (m s $^{-1}$)	5.16	3.82	3.95 $\pm$ 0.04	4.20 $\pm$ 0.23	4.23 $\pm$ 0.18	4.22 $\pm$ 0.03	3.36$\pm$0.05	3.59 $\pm$ 0.05	3.95 $\pm$ <0.01	3.55 $\pm$ 0.04
	ASR (W m $^{-2}$)	45.6	28.6	28.0 $\pm$ 0.4	67.0 $\pm$ 7.4	30.5 $\pm$ 0.5	26.0 $\pm$ 0.4	36.3 $\pm$ 1.3	32.1 $\pm$ 2.4	24.4$\pm$<0.1	25.1 $\pm$ 0.4
	OLR (W m $^{-2}$)	33.7	20.4	20.0 $\pm$ 0.3	26.3 $\pm$ 1.2	19.0 $\pm$ 0.3	19.0 $\pm$ 0.3	17.7 $\pm$ 0.5	17.1 $\pm$ 0.4	16.2$\pm$<0.1	16.3 $\pm$ 0.2
	Geometric mean	1.000	0.646	0.610 $\pm$ 0.008	0.755 $\pm$ 0.026	0.630 $\pm$ 0.015	0.609 $\pm$ 0.004	0.561 $\pm$ 0.007	0.560 $\pm$ 0.010	0.538 $\pm$ <0.001	0.518$\pm$0.002
	Geometric mean	1.000	0.646	0.610 $\pm$ 0.008	0.755 $\pm$ 0.026	0.630 $\pm$ 0.015	0.609 $\pm$ 0.004	0.561 $\pm$ 0.007	0.560 $\pm$ 0.010	0.538 $\pm$ <0.001	0.518$\pm$0.002

Table 12: Relative RMSE by variable under the shared-planets protocol, reported as mean $\pm$ standard deviation over random seeds. Deterministic baselines are shown without standard deviations. The main-text Table 2 shows only the means.

Sub-set	Variable	Train-mean	kNN	PCA-GBT	Coord-MLP	Coord-DeepONet	PCA-MLP	ConvDec	SFNO	PPCA-ICM	GPLFR
Multi-partial	Surface temp.	1.680	1.283	1.039 $\pm$ 0.113	1.380 $\pm$ 0.180	1.005 $\pm$ 0.113	1.057 $\pm$ 0.059	1.007 $\pm$ 0.048	0.975 $\pm$ 0.072	0.927 $\pm$ 0.002	0.739$\pm$0.005
	Temperature	1.655	1.209	0.925 $\pm$ 0.037	0.998 $\pm$ 0.073	0.949 $\pm$ 0.079	0.902 $\pm$ 0.033	0.903 $\pm$ 0.030	0.921 $\pm$ 0.029	0.829 $\pm$ 0.002	0.664$\pm$0.008
	Humidity	1.515	1.247	0.896 $\pm$ 0.032	1.008 $\pm$ 0.117	0.879 $\pm$ 0.100	0.818 $\pm$ 0.038	0.851 $\pm$ 0.011	0.824 $\pm$ 0.012	0.787 $\pm$ 0.008	0.670$\pm$0.004
	Cloud fraction	0.867	0.752	0.644 $\pm$ 0.004	0.733 $\pm$ 0.015	0.653 $\pm$ 0.017	0.639 $\pm$ 0.004	0.595 $\pm$ 0.008	0.606 $\pm$ 0.008	0.624 $\pm$ 0.004	0.562$\pm$0.004
	E-W wind	1.265	0.832	0.897 $\pm$ 0.012	0.937 $\pm$ 0.010	0.918 $\pm$ 0.039	0.909 $\pm$ 0.019	0.757$\pm$0.013	0.833 $\pm$ 0.028	0.785 $\pm$ 0.001	0.765 $\pm$ 0.007
	N-S wind	1.205	0.828	0.885 $\pm$ 0.015	0.898 $\pm$ 0.038	0.919 $\pm$ 0.022	0.837 $\pm$ 0.009	0.714$\pm$0.010	0.746 $\pm$ 0.014	0.802 $\pm$ <0.001	0.749 $\pm$ 0.005
	ASR	1.342	0.884	0.699 $\pm$ 0.015	2.007 $\pm$ 0.237	0.794 $\pm$ 0.042	0.699 $\pm$ 0.027	1.044 $\pm$ 0.030	0.920 $\pm$ 0.047	0.711 $\pm$ 0.002	0.594$\pm$0.011
	OLR	1.571	1.023	0.775 $\pm$ 0.023	1.218 $\pm$ 0.126	0.776 $\pm$ 0.022	0.767 $\pm$ 0.034	0.741 $\pm$ 0.014	0.765 $\pm$ 0.024	0.776 $\pm$ 0.004	0.640$\pm$0.007
	Geometric mean	1.360	0.988	0.836 $\pm$ 0.019	1.094 $\pm$ 0.029	0.853 $\pm$ 0.031	0.819 $\pm$ 0.016	0.813 $\pm$ 0.007	0.815 $\pm$ 0.010	0.776 $\pm$ 0.001	0.669$\pm$0.003
	Geometric mean	1.360	0.988	0.836 $\pm$ 0.019	1.094 $\pm$ 0.029	0.853 $\pm$ 0.031	0.819 $\pm$ 0.016	0.813 $\pm$ 0.007	0.815 $\pm$ 0.010	0.776 $\pm$ 0.001	0.669$\pm$0.003
Multi-complete	Surface temp.	1.709	1.445	1.011 $\pm$ 0.071	1.298 $\pm$ 0.098	1.009 $\pm$ 0.104	1.009 $\pm$ 0.049	1.026 $\pm$ 0.050	0.979 $\pm$ 0.130	1.039 $\pm$ 0.001	0.788$\pm$0.013
	Temperature	1.558	1.317	0.874 $\pm$ 0.020	0.947 $\pm$ 0.141	0.912 $\pm$ 0.095	0.850 $\pm$ 0.059	0.888 $\pm$ 0.017	0.924 $\pm$ 0.044	0.935 $\pm$ 0.001	0.687$\pm$0.007
	Humidity	1.534	1.274	0.837 $\pm$ 0.018	1.042 $\pm$ 0.100	0.857 $\pm$ 0.046	0.825 $\pm$ 0.041	0.874 $\pm$ 0.035	0.863 $\pm$ 0.074	0.853 $\pm$ <0.001	0.694$\pm$0.006
	Cloud fraction	0.861	0.748	0.635 $\pm$ 0.004	0.721 $\pm$ 0.017	0.665 $\pm$ 0.014	0.623 $\pm$ 0.005	0.605 $\pm$ 0.009	0.610 $\pm$ 0.016	0.615 $\pm$ <0.001	0.566$\pm$0.001
	E-W wind	1.377	0.974	0.863 $\pm$ 0.018	0.948 $\pm$ 0.020	0.956 $\pm$ 0.035	0.940 $\pm$ 0.017	0.801 $\pm$ 0.018	0.855 $\pm$ 0.035	0.854 $\pm$ <0.001	0.792$\pm$0.012
	N-S wind	1.164	0.930	0.851 $\pm$ 0.016	0.938 $\pm$ 0.011	0.963 $\pm$ 0.052	0.914 $\pm$ 0.010	0.767$\pm$0.015	0.773 $\pm$ 0.021	0.812 $\pm$ <0.001	0.779 $\pm$ 0.009
	ASR	1.355	0.853	0.698 $\pm$ 0.017	1.808 $\pm$ 0.238	0.809 $\pm$ 0.050	0.686 $\pm$ 0.016	1.000 $\pm$ 0.065	0.873 $\pm$ 0.062	0.737 $\pm$ <0.001	0.600$\pm$0.004
	OLR	1.579	0.974	0.763 $\pm$ 0.011	1.087 $\pm$ 0.044	0.782 $\pm$ 0.048	0.761 $\pm$ 0.018	0.741 $\pm$ 0.014	0.767 $\pm$ 0.044	0.797 $\pm$ 0.001	0.641$\pm$0.006
	Geometric mean	1.365	1.040	0.809 $\pm$ 0.014	1.059 $\pm$ 0.049	0.861 $\pm$ 0.036	0.816 $\pm$ 0.019	0.827 $\pm$ 0.015	0.822 $\pm$ 0.029	0.822 $\pm$ <0.001	0.688$\pm$0.006
	Geometric mean	1.365	1.040	0.809 $\pm$ 0.014	1.059 $\pm$ 0.049	0.861 $\pm$ 0.036	0.816 $\pm$ 0.019	0.827 $\pm$ 0.015	0.822 $\pm$ 0.029	0.822 $\pm$ <0.001	0.688$\pm$0.006

Table 13: RMSE on the Multi-partial subset for predictions from training seed 0, reported with 95% bootstrap intervals from 1,000 resamples of the 128 test examples. Each resample draws 128 examples with replacement and is applied to every method jointly. Lower is better.

Variable	Train-mean	kNN	PCA-GBT	PCA-MLP	Coord-MLP	Coord-DeepONet	ConvDec	SFNO	PPCA-ICM	GPLFR
Surface temp. (K)	24.52 ^{+3.55} _{-3.08}	18.79 ^{+3.15} _{-2.72}	11.87 ^{+1.74} _{-1.57}	12.4 ^{+1.83} _{-1.71}	17.18 ^{+1.84} _{-1.72}	12.8 ^{+2.27} _{-2.07}	12.6 ^{+1.62} _{-1.46}	13.2 ^{+1.85} _{-1.66}	10.09 ^{+1.75} _{-1.55}	9.83 ^{+1.59} _{-1.50}
Temperature (K)	20.34 ^{+2.46} _{-2.07}	14.66 ^{+2.25} _{-1.90}	10.19 ^{+1.44} _{-1.30}	10.1 ^{+1.48} _{-1.22}	12.65 ^{+1.26} _{-1.14}	10.7 ^{+1.59} _{-1.42}	10 ^{+1.17} _{-1.1}	10.8 ^{+1.36} _{-1.22}	8.82 ^{+1.39} _{-1.21}	8.19 ^{+1.27} _{-1.04}
Humidity (dex)	1.119 ^{+0.108} _{-0.101}	0.784 ^{+0.114} _{-0.096}	0.537 ^{+0.074} _{-0.060}	0.528 ^{+0.089} _{-0.068}	0.657 ^{+0.060} _{-0.056}	0.53 ^{+0.076} _{-0.071}	0.515 ^{+0.062} _{-0.059}	0.531 ^{+0.072} _{-0.0618}	0.443 ^{+0.072} _{-0.061}	0.429 ^{+0.071} _{-0.059}
Cloud fraction (1)	0.1277 ^{+0.0059} _{-0.0058}	0.0998 ^{+0.0088} _{-0.0081}	0.0888 ^{+0.0054} _{-0.0056}	0.0903 ^{+0.0061} _{-0.0062}	0.1051 ^{+0.049} _{-0.040}	0.0907 ^{+0.0633} _{-0.0599}	0.0822 ^{+0.0011} _{-0.0017}	0.0851 ^{+0.00579} _{-0.00579}	0.0839 ^{+0.0071} _{-0.0071}	0.0774 ^{+0.0056} _{-0.0054}
E-W wind (m s ⁻¹)	16.73 ^{+1.28} _{-1.24}	10.92 ^{+1.36} _{-1.31}	10.68 ^{+1.15} _{-0.96}	11.2 ^{+1.14} _{-1.1}	12.22 ^{+0.99} _{-0.99}	10.9 ^{+1.2} _{-1.15}	9.9 ^{+1.17} _{-1.17}	10.6 ^{+1.07} _{-1.01}	9.63 ^{+1.20} _{-1.17}	9.45 ^{+1.12} _{-1.01}
N-S wind (m s ⁻¹)	6.57 ^{+0.41} _{-0.41}	4.72 ^{+0.50} _{-0.51}	4.81 ^{+0.41} _{-0.47}	4.65 ^{+0.396} _{-0.354}	4.96 ^{+0.38} _{-0.38}	4.99 ^{+0.568} _{-0.48}	3.91 ^{+0.408} _{-0.39}	4.28 ^{+0.375} _{-0.372}	4.33 ^{+0.42} _{-0.41}	4.12 ^{+0.38} _{-0.37}
ASR (W m ⁻²)	59.0 ^{+5.0} _{-5.2}	33.2 ^{+4.8} _{-4.5}	27.2 ^{+4.2} _{-3.6}	26.9 ^{+4.05} _{-3.46}	77.1 ^{+4.5} _{-4.1}	29.7 ^{+4.17} _{-3.97}	37.1 ^{+3.92} _{-3.65}	32.4 ^{+3.98} _{-3.69}	26.0 ^{+4.2} _{-3.7}	22.5 ^{+4.1} _{-3.6}
OLR (W m ⁻²)	41.0 ^{+3.0} _{-2.8}	24.8 ^{+3.0} _{-3.0}	19.2 ^{+3.1} _{-2.1}	19.1 ^{+3.18} _{-2.48}	34.3 ^{+3.5} _{-2.5}	19 ^{+3.2} _{-2.2}	17.4 ^{+3.66} _{-3.65}	18.7 ^{+2.6} _{-2.6}	18.0 ^{+2.2} _{-2.2}	16.0 ^{+2.2} _{-2.2}

Table 14: Paired bootstrap test of the seed-0 Multi-partial RMSE ranking. Each row compares adjacent methods using the same 1,000 joint resamples as Table 13. We show the ratio of the lower-ranked to the higher-ranked method’s geometric-mean RMSE, its 95% interval, and the fraction of resamples in which the ordering holds.

Higher-ranked	Lower-ranked	Ratio	95% interval	Holds (%)
GPLFR	PPCA-ICM	1.070	[1.018, 1.128]	99.5
PPCA-ICM	ConvDec	1.095	[1.035, 1.159]	100.0
ConvDec	PCA-GBT	1.018	[0.972, 1.062]	77.5
PCA-GBT	PCA-MLP	1.005	[0.959, 1.059]	58.0
PCA-MLP	SFNO	1.012	[0.976, 1.049]	72.9
SFNO	Coord-DeepONet	1.018	[0.972, 1.065]	77.2
Coord-DeepONet	kNN	1.207	[1.102, 1.330]	100.0
kNN	Coord-MLP	1.128	[1.027, 1.234]	99.8

high ACC for ASR, reflecting the strong dependence of this field on substellar point geometry, which is consistent across these tidally locked planets.

Table 15: Anomaly correlation coefficient (ACC) by variable and dataset subset. Higher is better; maximum is 1. Values are the mean of five training seeds (one deterministic run for Train-mean and kNN).

Sub-set	Variable	Train-mean	kNN	PCA-GBT	Coord-MLP	Coord-DeepONet	PCA-MLP	ConvDec	SFNO	PPCA-ICM	GPLFR
Multi-partial	Surface temp.	0.941	0.968	0.963	0.907	0.947	0.946	0.927	0.955	0.971	0.973
	Temperature	0.400	0.634	0.603	0.405	0.574	0.631	0.490	0.524	0.629	0.687
	Humidity	0.682	0.780	0.753	0.656	0.732	0.761	0.666	0.708	0.775	0.792
	Cloud fraction	0.535	0.635	0.664	0.568	0.665	0.676	0.690	0.687	0.691	0.712
	E-W wind	0.528	0.670	0.654	0.583	0.646	0.666	0.679	0.687	0.696	0.728
	N-S wind	0.544	0.665	0.646	0.609	0.627	0.655	0.688	0.689	0.672	0.699
	ASR	0.994	0.995	0.996	0.969	0.995	0.996	0.993	0.995	0.996	0.997
	OLR	0.712	0.815	0.855	0.706	0.857	0.853	0.868	0.857	0.871	0.888
Multi-complete	Surface temp.	0.941	0.965	0.962	0.918	0.933	0.938	0.933	0.953	0.973	0.972
	Temperature	0.422	0.634	0.645	0.477	0.590	0.657	0.508	0.551	0.663	0.700
	Humidity	0.727	0.814	0.810	0.708	0.783	0.804	0.722	0.762	0.835	0.848
	Cloud fraction	0.579	0.678	0.719	0.643	0.702	0.725	0.730	0.732	0.736	0.756
	E-W wind	0.537	0.657	0.683	0.606	0.674	0.672	0.705	0.710	0.723	0.742
	N-S wind	0.555	0.647	0.654	0.632	0.646	0.650	0.698	0.691	0.684	0.703
	ASR	0.994	0.995	0.996	0.973	0.995	0.996	0.993	0.995	0.996	0.997
	OLR	0.714	0.821	0.854	0.741	0.852	0.844	0.869	0.863	0.865	0.893
Single-complete	Surface temp.	0.955	0.973	0.971	0.948	0.954	0.955	0.964	0.973	0.971	0.978
	Temperature	0.448	0.685	0.611	0.517	0.581	0.628	0.571	0.598	0.680	0.692
	Humidity	0.738	0.832	0.822	0.744	0.793	0.798	0.778	0.801	0.823	0.848
	Cloud fraction	0.599	0.706	0.719	0.652	0.708	0.701	0.732	0.722	0.724	0.749
	E-W wind	0.510	0.717	0.676	0.621	0.673	0.668	0.732	0.712	0.708	0.741
	N-S wind	0.566	0.706	0.662	0.620	0.666	0.637	0.721	0.690	0.658	0.699
	ASR	0.993	0.995	0.996	0.973	0.995	0.996	0.993	0.995	0.996	0.996
	OLR	0.797	0.882	0.863	0.776	0.871	0.843	0.912	0.899	0.897	0.911

F.4 Probabilistic evaluation results

PPCA-ICM and GPLFR are our two naturally probabilistic baselines, and the best-performing ones by RMSE. Here we show their energy scores (Table 16) and spread-skill ratios (Table 17) on the three subsets.

Energy score. The energy score is

$$\text{ES}(p, \mathbf{y}) = \mathbb{E}_{\mathbf{y}' \sim p} \|\mathbf{y}' - \mathbf{y}\|_G - \frac{1}{2} \mathbb{E}_{\mathbf{y}', \mathbf{y}'' \sim p} \|\mathbf{y}' - \mathbf{y}''\|_G,$$

where p is the predictive distribution, $\mathbf{y}$ is the ground-truth output, and $\|\mathbf{e}\|_G = \sqrt{\mathbf{e}^\top \mathbf{G} \mathbf{e}}$ with $\mathbf{G}$ containing latitude area weights. The first term penalizes each draw’s distance from the truth, which grows with both the predictive distribution’s misplacement and its spread; the second rewards spread,

Table 16: Energy score by variable and subset. Lower is better. Scores are the mean of five training seeds.

Variable	Multi-partial		Multi-complete		Single-complete	
	PPCA-ICM	GPLFR	PPCA-ICM	GPLFR	PPCA-ICM	GPLFR
Surface temp. (K)	7.82	7.98	8.75	8.03	8.34	7.57
Temperature (K)	6.89	6.68	7.64	6.67	7.18	6.41
Humidity (dex)	0.345	0.358	0.355	0.341	0.399	0.354
Cloud fraction (1)	0.0674	0.0586	0.0707	0.0623	0.0725	0.0566
E-W wind (m s^{-1})	8.02	7.31	7.27	6.36	7.31	5.93
N-S wind (m s^{-1})	3.58	3.17	3.48	3.12	3.59	2.77
ASR (W m^{-2})	20.7	18.7	20.8	18.3	21.9	19.8
OLR (W m^{-2})	14.2	12.7	14.2	12.3	14.2	12.9

Table 17: Spread-skill ratio (SSR) by variable and subset. Closer to one is better. Values are the mean of five training seeds.

Variable	Multi-partial		Multi-complete		Single-complete	
	PPCA-ICM	GPLFR	PPCA-ICM	GPLFR	PPCA-ICM	GPLFR
Surface temp.	0.686	1.091	0.628	1.113	0.241	0.887
Temperature	0.626	1.031	0.541	0.996	0.239	0.911
Humidity	0.741	1.166	0.694	1.214	0.205	0.847
Cloud fraction	0.394	1.138	0.377	1.199	0.181	0.926
E-W wind	0.305	0.678	0.285	0.682	0.174	0.764
N-S wind	0.374	0.853	0.341	0.837	0.155	0.717
ASR	0.393	1.041	0.401	1.104	0.145	0.678
OLR	0.474	1.087	0.489	1.139	0.204	0.966

at a rate balanced such that under- and overconfidence both worsen the expected score; lower is better. We estimate the energy score using 64 posterior predictive samples. Its results largely follow the RMSE results, with GPLFR beating PPCA-ICM on most variables and subsets (22 of the 24 variable-subset combinations) (Table 16).

Spread-skill ratio. SSR is a first-order diagnostic of ensemble calibration. We define it as

$$\text{SSR} = \sqrt{\frac{\sum_i \text{Spread}_i^2}{\sum_i \text{MSE}_i}},$$

where for each test example i ,

$$\text{Spread}_i^2 = \frac{1}{M-1} \sum_{m=1}^M \|\mathbf{y}_i^{[m]} - \hat{\mathbf{y}}_i\|_G^2, \quad \text{MSE}_i = \|\hat{\mathbf{y}}_i - \mathbf{y}_i\|_G^2,$$

and $\hat{\mathbf{y}}_i$ is the ensemble mean. A well-calibrated ensemble has $\text{SSR} \approx 1$; values below 1 indicate overconfidence (too little spread) and values above 1 indicate underconfidence.

On SSR, PPCA-ICM is consistently overconfident, while GPLFR is more balanced between over- and underconfidence, leaning overconfident on Single-complete. (Table 17). Both models improve going from Single-complete to the Multi- subsets. By this measure, GPLFR is better calibrated than PPCA-ICM on all variables and subsets.

F.5 Relative energy score results

Table 18 reports relative energy scores for the probabilistic methods (PPCA-ICM, GPLFR) under the shared-planets protocol (Section 2.6). Following the same ratio structure as the relative RMSE, we define

$$\text{ES}_{\text{rel}} = \frac{\text{ES}(p, \mathbf{y}^s)}{\|\mathbf{y}^{s'} - \mathbf{y}^s\|_G},$$

Table 18: Relative energy score by variable under the shared-planets protocol for probabilistic methods. Bold indicates lowest (best) score within each subset. Scores are the mean of five training seeds.

Variable	Multi-partial		Multi-complete	
	PPCA-ICM	GPLFR	PPCA-ICM	GPLFR
Surface temp.	0.712	0.623	0.802	0.673
Temperature	0.638	0.551	0.724	0.578
Humidity	0.590	0.563	0.639	0.591
Cloud fraction	0.505	0.421	0.495	0.430
E-W wind	0.636	0.581	0.702	0.605
N-S wind	0.653	0.585	0.657	0.602
ASR	0.562	0.476	0.582	0.487
OLR	0.607	0.504	0.620	0.511
Geometric mean	0.610	0.534	0.647	0.555

where the denominator is the other GCM’s energy score, which reduces to the RMSE because GCMs only produce point predictions (i.e., delta-function predictive distributions). Probabilistic emulators that produce well-calibrated spreads can therefore achieve lower relative energy scores than relative RMSEs, reflecting the additional information in a calibrated predictive distribution over a point estimate.

Both methods achieve relative energy score below 1 on every variable in both Multi- subsets. GPLFR’s geometric mean relative energy score is 0.534 on Multi-partial, 20% lower than its geometric mean relative RMSE (0.669; Table 2), indicating the value of a distributional prediction over a point prediction (since both metrics share a denominator). PPCA-ICM shows a similar drop of about 21%; the Multi-complete drops are 19% and 21%.

F.6 Per-example relative RMSE results

Figure 3 shows the distribution of per-example relative RMSEs for GPLFR on the Multi-partial dataset. While the medians are well below 1 for all variable groups (ranging from 0.54 for ASR to 0.68 for N–S wind), the spread is significant. Surface temperature has the widest distribution, with two examples exceeding a relative RMSE of 2 (2.1 and 5.0 respectively), meaning the emulator’s error is far larger than GCM disagreement for those examples. The two worst examples overall by geometric-mean relative RMSE (simulations 28 and 1581, at 1.55 and 1.51; use `simulation_id` in the code/data) are both ExoCAM cases, consistent with it having fewer training data than the UM.

The high relative RMSE tails suggest that certain (planet, GCM) combinations remain difficult to emulate reliably and represent natural targets for future method development or additional training data. We show predictions for these two outliers alongside two more typical examples (simulations 16 and 1629) in Figures 4, 5, and 6.

G Example predictions and climate diagnostics

Here we show some ground truth and baseline predictions using common plots and climate diagnostics from exoplanet science. Figures 4, 5, and 6 show plots for four test examples: 28 and 1581 are two of the worst predicted by our strongest baseline (GPLFR); the other two are more typical predictions. Figure 7 shows six scalar climate diagnostics for all planets in the Multi-partial test set.

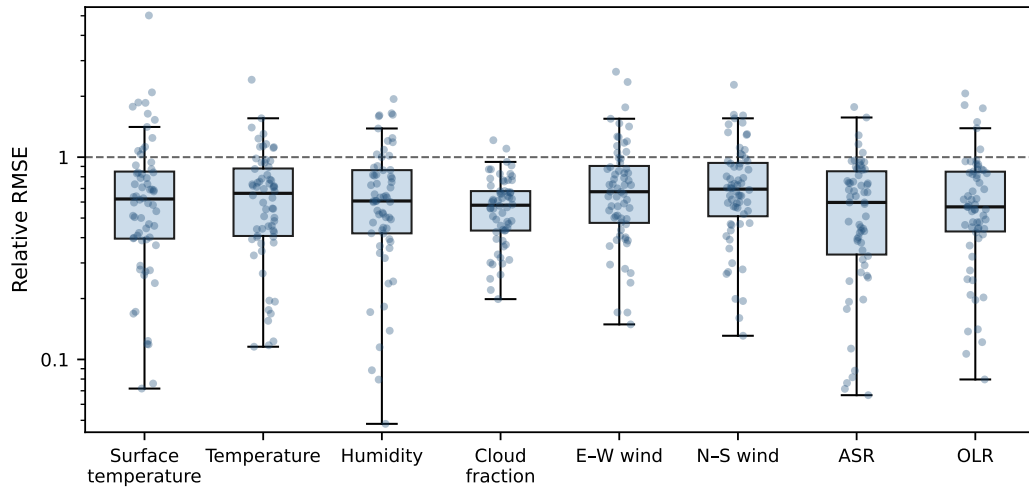


Figure 3: Per-example relative RMSE distribution for GPLFR on Multi-partial, grouped by variable. Each point is one example ((planet, GCM) combination). The dashed line at 1 marks the GCM-disagreement threshold. Boxes show the interquartile range; whiskers extend to $1.5 \times \text{IQR}$. Scores are from training seed 0.

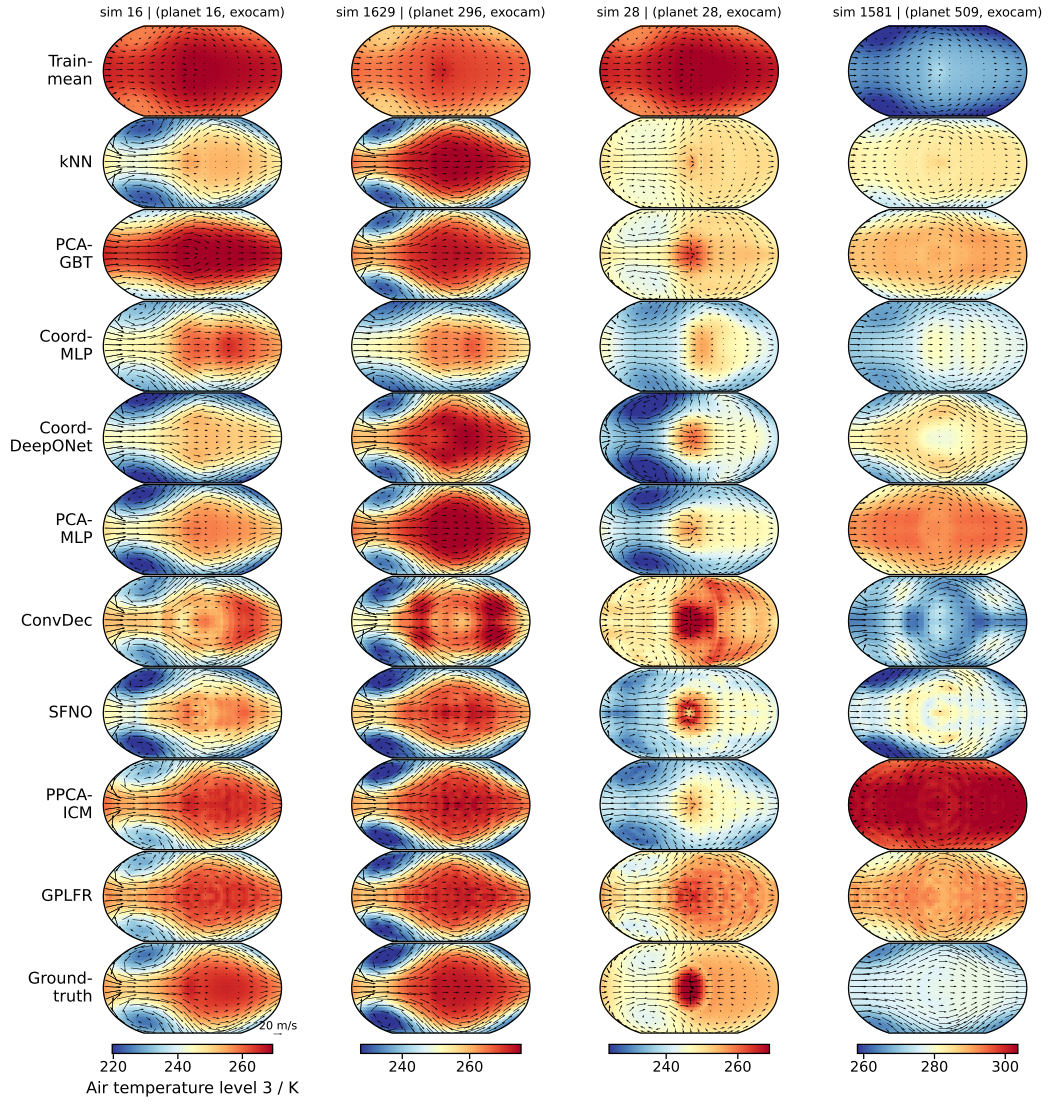


Figure 4: Spatial maps of temperature with superimposed wind vectors at relative isobar $\sigma_3 \approx 0.72$ (this would be around the mid-troposphere on Earth) for four test examples. Simulations 28 and 1581 are the two worst predicted by our strongest baseline (GPLFR), while simulations 16 and 1629 give more typical predictions.

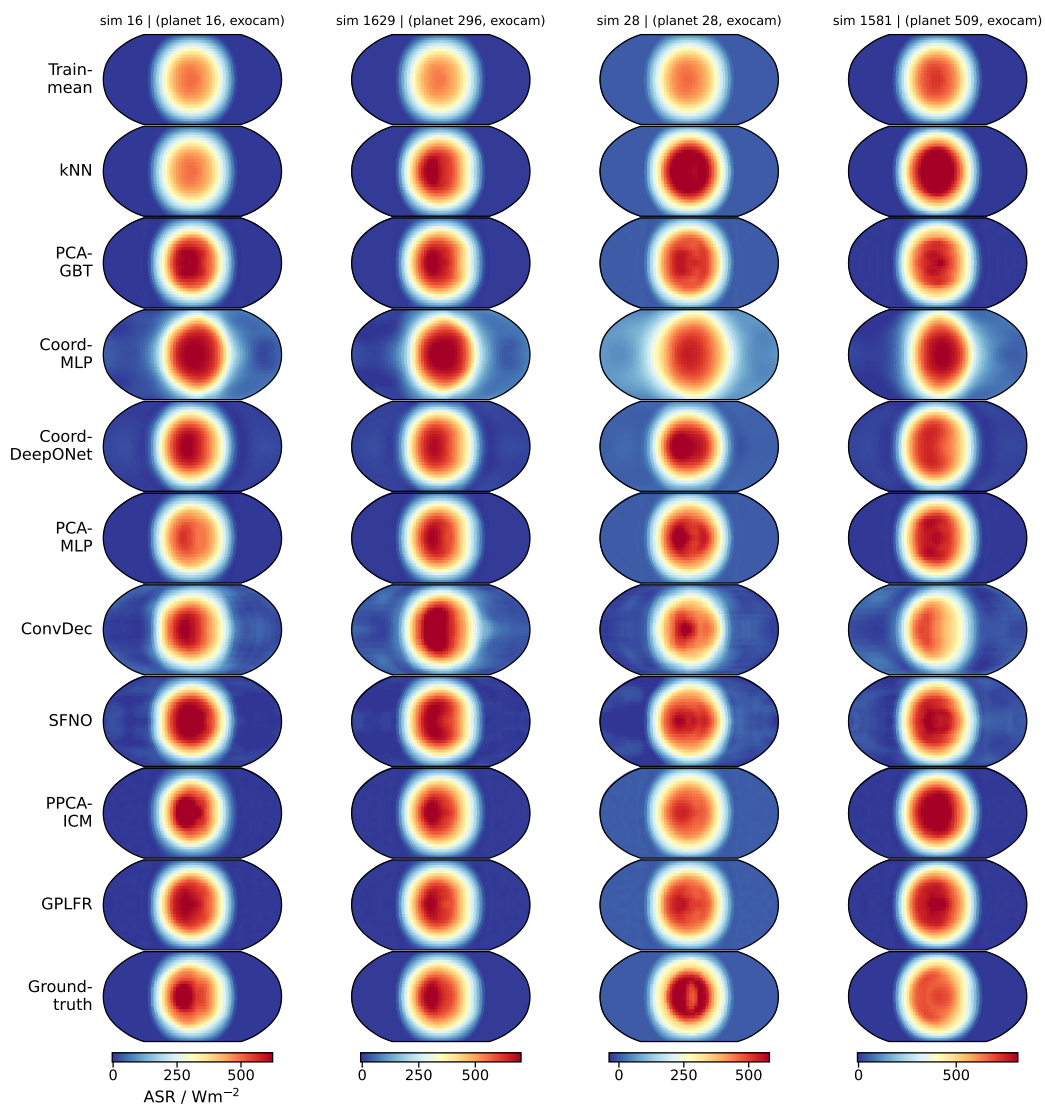


Figure 5: Absorbed shortwave radiation (ASR) maps for four test examples. Simulations 28 and 1581 are the two worst predicted by our strongest baseline (GPLFR), while simulations 16 and 1629 give more typical predictions.

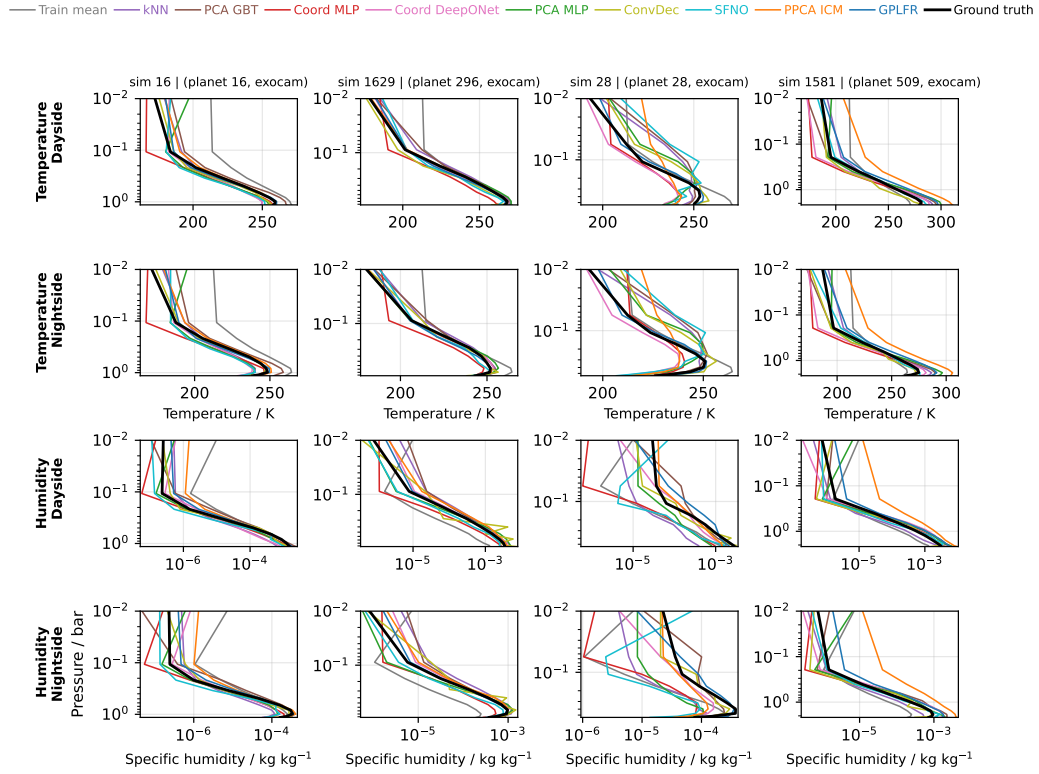


Figure 6: Dayside and nightside vertical profiles of area-weighted mean temperature and specific humidity for four test examples. Simulations 28 and 1581 are the two worst predicted by our strongest baseline (GPLFR), while simulations 16 and 1629 give more typical predictions.

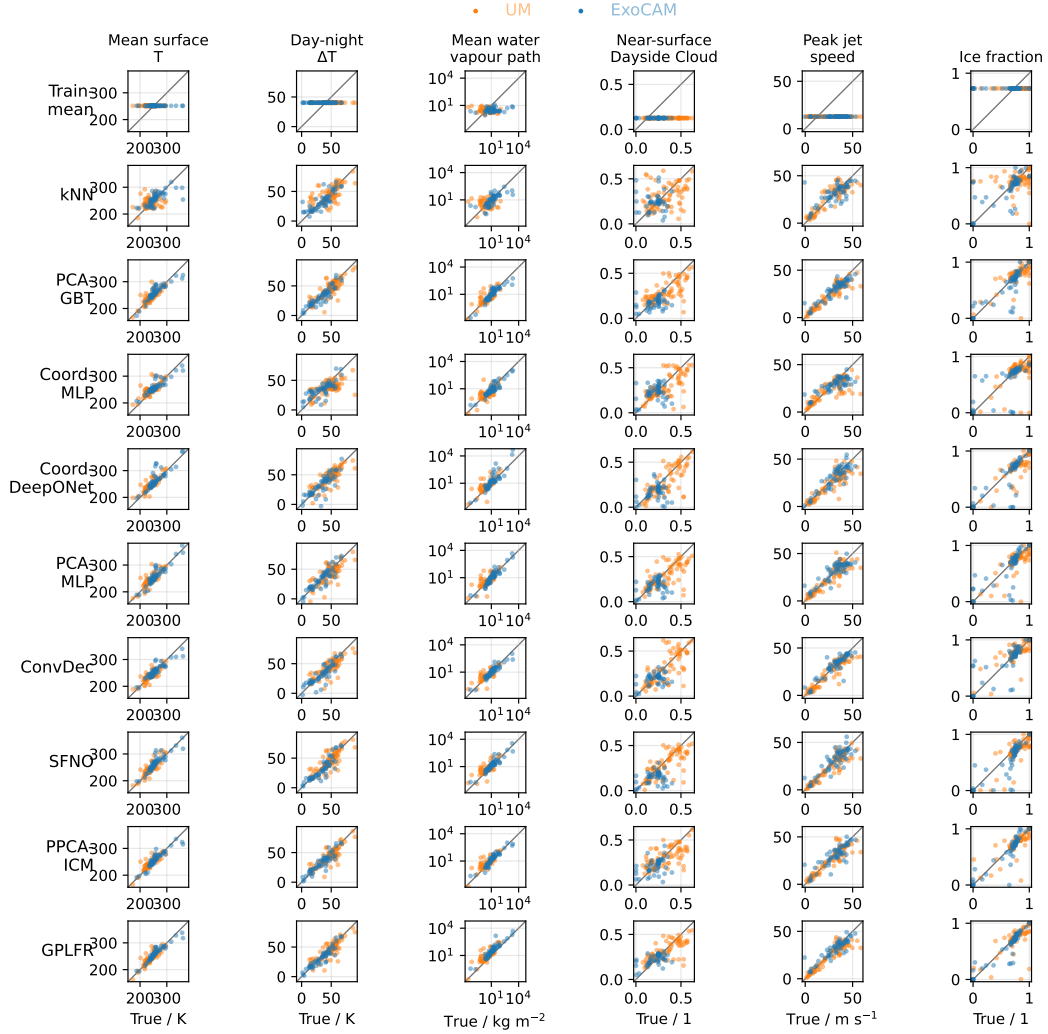


Figure 7: Predicted (y-axis) versus true (x-axis) values for six climate diagnostics across the Multi-partial test set. Each point is one test example; the line marks perfect prediction. Points are coloured by the source GCM. Mean surface T : area-weighted global-mean surface temperature. Day-night ΔT : area-weighted mean dayside minus nightside surface temperature. Water vapour path: total mass of water vapour in an atmospheric column, integrated over pressure levels and area-weighted over the globe. Near-surface dayside cloud: cloud fraction at the lowest pressure level, averaged over dayside grid cells. Peak jet speed: maximum of the longitudinally averaged E-W wind over latitudes and pressure levels. Ice fraction: fraction of surface area with $T_{\text{surf}} < 273 \text{ K}$.